

ENT 221

Agripreneurship

Study Manual

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Fish farming



Oil palm



Value chain



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Every line is a link. The page numbers are read back from the rendered book, never typed by hand.

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How to use this manual

Read this page once. It is short, and it decides whether the rest of the manual works for you.

TEACH

THE BARGAIN

This manual assumes you know nothing about fish farming, oil palm, or agribusiness beyond having eaten food. It carries you from there to a full score in the computer-based test. It is meant to be **enough on its own**: you should never need the lecturer, the course text, or a website to follow it. Anything it uses, it teaches first.

In exchange it asks one thing: **test yourself before you look**. Reading a fact and recognising it later is not the same as being able to produce it when a question hides the answer among four near-identical options. Every box that asks you something hides its answer under a line. Cover it, answer out loud or on paper, and only then check.

The boxes

Every page is built from a small set of boxes. Each has one job, and the colour tells you which before you read a word.

TEACH

New ideas, explained from nothing. Read these properly.

● MUST MEMORISE

Harvest gold means **learn this by heart**. The definitions the examiner asks for word for word, and above all the **enumerated lists**: the three water parameters, the five fish-farm departments, the 3Ms and the ten MAKE principles, the five stages of palm oil processing, the six types of value addition. It is the only thing this colour is ever used for. If a page has gold on it, there is something on it you are expected to know cold.

WORKED

SOLVED WHERE IT IS TAUGHT

A real test question, solved in full, then a **now you try**. Each worked box carries a small chip saying where the question comes from, so you know it was actually asked:

TEST 2 Q1 a real class test (with its confirmed answer); **TEXT 2.1.4** the course text's own section. Do the "now you try" before you move on.

⚠ TRAP

A mistake that is easy to make and costly. Most are taken from the exact snares the examiner sets: an option that says "6.5 to 8.5" as the pH range slipped into the dissolved oxygen question, a distractor that sounds thorough because it says "remove all bacteria", the difference between the fruit **type** *Nigrescens* and the fruit **species** *Dura*.

RECALL AND CHECK

A question with its answer printed under a dashed line. **Cover the answer.** Say or write yours first, then look. Checking whether you knew something is not the same as knowing it, and the difference is the whole point of this box.

FILL IN THE GAP

A sentence with a word removed, exactly the way the test's fill-in-the-gap questions are built. The missing word is almost always a single term lifted from a bulleted list or a bracket in the text. The answers sit below the box.

AS PRINTED

A real test question, reproduced exactly as it appeared on screen, in this dark As printed bar, so you meet it the way the examiner set it.

A violet **Break it down** box then reads it in your shoes, and a teal **Answer** box works it in full and names the correct option.

One promise, and one label

TEACH

The promise. Every question quoted in this manual was transcribed by eye from the real test screenshots, and every answer worked out independently, not copied. Where the test itself reaches outside the course text or paraphrases it loosely, this manual quotes the question exactly and flags it, because in the test you answer what is on the screen.

The label. Almost everything examined is in the course text. The rare item that is not, or that the test words differently from the text, carries a **NOT IN THE COURSE TEXT** label. It is still taught in full, because it can still be asked; the label only tells you that going back to the text alone will not find it. This manual is where you find it.

BEFORE YOU START

The Exam at a Glance

What the test looks like and where the marks live. Learn this before you learn anything else: it tells you where to aim.

Assessment in this course is by **computer-based test (CBT)**. Each question is worth **one mark**, and there is no penalty structure to game: every question is winnable if you have read the material. The two class tests were **thirty questions** each, sat in about twenty minutes, so the pace is brisk. Expect the final assessment in the same style.

Four question styles appear, and this manual drills all four:

- ▶ **Multiple choice** (single best answer, options A to D): the bulk of the paper.
- ▶ **True or false**: a single statement to judge.
- ▶ **Fill in the gap**: type the one missing word.
- ▶ **Matching**: pair each item on the left with its partner on the right.

● MUST MEMORISE

THE TOPIC MAP

The marks come from four modules. Learn this map; each item is a module in this manual.

- ▶ **Module One, Agriculture and Agripreneurship**: what agriculture and an agripreneur are, the value chain and its segments, the importance of agripreneurship.
- ▶ **Module Two, Fish Farming**: the departments, investment potential, the three water parameters, fish behaviour, the 3M wealth model, and marketing. **The most heavily tested.**
- ▶ **Module Three, Oil Palm**: cultivation (varieties, spacing, agronomy, harvest) and processing (the five stages, by-products).
- ▶ **Module Four, Agribusiness Opportunities and Value Addition**: opportunity criteria, the value chain, value addition types and forms, and production and operations management.

⚠ TRAP

DO NOT REVISE ONLY FISH

Both class tests leaned almost entirely on Modules One and Two, and fish farming dominated. It is tempting to conclude that oil palm and value addition do not matter. Do not. The final assessment can draw on any module, and the facts in Modules Three and Four (the Tenera hybrid, the five processing stages, the six types of value addition, the POCD framework) are exactly the kind of crisp, list-based content a CBT loves. This manual teaches and tests all four modules to the same depth for that reason.

BEFORE YOU START

How the Examiner Thinks

A CBT is not a wall of random facts. This one has habits, and once you can read them you can often find the answer even on a fact you half-remember. Learn these tells.

- ▶ **The questions come straight from the text.** Almost every item traces to a specific line, often word for word. This rewards reading the material, not clever reasoning about agriculture in general. If an option uses wording you recognise from a bullet in the text, it is usually the answer.
- ▶ **Absolute words mark a wrong option.** Distractors love **every, all, entirely, solely, exclusively, only, always**. "Remove all bacteria", "operates entirely as a non-profit", "convert every farm exclusively": these are built to be wrong. The correct answer is the **measured** one.
- ▶ **The correct answer is often the longest, most specific option.** The examiner builds the right answer by compressing a whole list from the text into one sentence, which makes it wordy. When two options look plausible, the fuller, more complete one usually wins.
- ▶ **Fill-in-the-gap answers are single words from lists and brackets.** hatchery, kitchen, art, grading, water, market, 8.5. To own these, memorise the **enumerated lists** and the **parenthetical asides**, which is exactly what the gold boxes in this manual are for.

△ TRAP

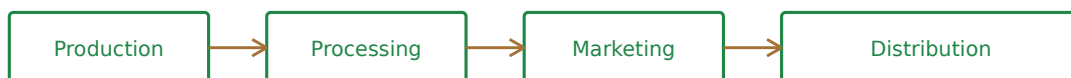
THE TELLS ARE AIDS, NOT LAWS

These habits are strong but not absolute, so never switch off your knowledge and answer by pattern alone. The test occasionally reaches outside the text or paraphrases it: one real question asked about a "total plate count", a phrase the text never uses (it says **Total Bacterial Count**), and another spoke of "water changes" where the text says **partial water changes**. Where this manual meets such a case it flags it, so you learn the fact *and* the loose wording. Use the tells to break a genuine tie, not to replace revision.

MODULE ONE

Agriculture and Agripreneurship

Before fish or oil palm, one idea underlies the whole course: farming stops being mere survival and becomes a business the moment you add value along a chain and sell it. This module defines agriculture, the agripreneur who works it as an enterprise, and the value chain every later module hangs on.



Understanding Agriculture and Agripreneurship

What agriculture is, how its parts connect, what turns a farmer into an agripreneur, and the value chain you can enter at any point to make money.

What agriculture is

TEACH

TEXT 1.1.1

Agriculture is broad. At its root every branch shares one thing: it works with **living organisms**, so there is always a seed time and a harvest time, and care in between decides the yield. The examiner's preferred one-line definition is worth learning exactly, because it has been used as a fill-in question.

● MUST MEMORISE

DEFINITION

Agriculture is the **science, art, and practice** of cultivating plants and raising animals for food, fibre, medicine, and other products used to sustain and enhance human life.

◆ **SCIENCE, ART, AND PRACTICE: THREE WORDS, AND "ART" IS THE ONE THE TEST PULLS OUT AS A GAP (IT REUSES THE SAME "SCIENCE AND ART" PHRASING FOR FISH FARMING).**

TEACH

THE ELEVEN ASPECTS

Agriculture spans eleven key aspects. You do not need to recite all eleven often, but two of them are favourite multiple-choice answers, so know what each *does*: a question describes the job and asks for the name.

- ▶ **Crop Production:** growing cereals, fruits, vegetables, cash crops.
- ▶ **Animal Husbandry:** rearing livestock such as cattle, poultry, sheep, and fish.
- ▶ **Soil Management:** maintaining fertility, preventing erosion.
- ▶ **Irrigation and Water Management:** efficient use of water.
- ▶ **Agricultural Engineering:** tools, machinery, infrastructure.
- ▶ **Agroforestry:** combining trees and crops.
- ▶ **Agricultural Economics:** the business side, trade and markets.
- ▶ **Food Science and Technology:** processing, preserving, improving food.
- ▶ **Sustainable and Organic Farming:** eco-friendly practices.
- ▶ **Plant Breeding:** developing new crop varieties to increase yield, resist pests, or tolerate the environment.
- ▶ **Pest and Disease Management:** controlling pests, diseases, and weeds.

Which specific branch of agriculture develops new crop varieties to increase yield, enhance pest resistance, or improve environmental tolerance?

- a Plant Breeding**
- b Soil Management
- c Agricultural Engineering
- d Agroforestry

ANSWER **a, Plant Breeding.** The verb "developing new crop varieties" is lifted straight from the text's definition. Soil Management keeps soil fertile; Agricultural Engineering builds tools; Agroforestry mixes trees and crops. None breeds varieties.

The aspects are not islands; modern agriculture is **interconnected sectors**. The textbook example, which the test uses almost verbatim, is livestock and crops: livestock provide **manure** that enriches the soil, crops and their residues serve as **fodder** for animals, and healthy soil then grows better crops. One good sentence covers it.

From farmer to agripreneur

Agripreneurship is simply **entrepreneurship in agriculture**: bringing a **business mindset** to agriculture, adding value to products or services so the sector becomes efficient, sustainable, and commercially viable.

An agripreneur identifies opportunities in agriculture and agribusiness and uses entrepreneurial skills to innovate, manage risk, and build profitable, sustainable ventures.

◆ **BRINGING A "BUSINESS" MINDSET TO AGRICULTURE: "BUSINESS" IS A CONFIRMED FILL-IN ANSWER.**

TEACH

AGRIPRENEUR VS TRADITIONAL FARMER

The single most repeated Module One idea is how an agripreneur differs from a traditional farmer. The agripreneur treats farming as a scalable **business**; the traditional farmer treats it as a way of life. Learn the contrast across a few dimensions.

DIMENSION	AGRIPRENEUR	TRADITIONAL FARMER
Mindset	Entrepreneurial; farming as a scalable business	Subsistence; farming as tradition or a way of life
Goal	Profit and scaling up	Meeting family needs or small local sales
Technology	Modern tools, apps, mechanisation, data	Mostly manual labour and inherited practice
Marketing	Branding, value addition, diverse markets	Sells to middlemen or the local market
Risk	Takes calculated risks, may insure	Avoids risk, relies on experience

PICK

PRACTICE · CF. TEST 2 Q2

Which option best distinguishes an agripreneurial venture from a traditional cultivation model?

- (a) Agripreneurship limits all business operations strictly to primary open-field crop cultivation and excludes digital technology.
- (b) Agripreneurship eliminates biological farming and focuses exclusively on speculative international financial trading.
- (c) **Agripreneurship applies entrepreneurial and business principles across agriculture and its value chain, including production, processing, distribution, technology and services.**
- (d) Agripreneurship operates entirely as a non-profit government programme and avoids commercial transactions, value addition and financial management.

ANSWER **c**. Notice the tells at work: a, b and d each carry an absolute (*strictly, exclusively, entirely*) and each is wrong. Option c is the longest and most complete, and it is the measured one. This is the examiner's signature.

The agricultural value chain

● MUST MEMORISE

DEFINITION

A value chain is the entire series of activities and transactions needed to make a product and deliver it to consumers.

◆ **LEARN THIS LINE WORD FOR WORD. IT APPEARS AS A FILL-IN ("...IS KNOWN AS A VALUE CHAIN") AND IS REUSED FOR THE FISH PRODUCT CHAIN IN MODULE TWO.**

TEACH

SEGMENTS YOU CAN ENTER

An agripreneur can invest in any segment of the chain. Two segments are favourite multiple-choice answers, so know their one-line jobs:

- ▶ **Production:** cultivating crops and rearing livestock to raw matured products.
- ▶ **Processing:** transforming raw matured products into value-added consumable or industrial goods, by **milling, smoking, canning, freezing**.
- ▶ **Marketing, Distribution, Agricultural Finance, Technology, Education and Training, Policy and Advocacy.**
- ▶ **Agricultural Waste Management:** composting and recycling agricultural waste into valuable products such as **biofuels and animal feeds**.
- ▶ **Infrastructure Development, Agricultural Tourism, Consulting.**

FILL IN THE GAP

PRACTICE · CF. TEST 1 Q6, Q12

1. The value-chain component that recycles agricultural waste into valuable products such as biofuels and animal feeds is _____.
2. Transforming raw matured products into value-added goods through milling, smoking, or canning is the _____ activity of the chain.

ANSWERS 1. Agricultural Waste Management 2. Processing.

● MUST MEMORISE

IMPORTANCE

Three benefits of agripreneurship, and the first is a confirmed true/false answer:

- ▶ **Economic Growth:** jobs, income, and innovation, especially in rural areas.
- ▶ **Food Security:** more and better-distributed food. ("Food security is a major benefit of agripreneurship" is **true**.)
- ▶ **Sustainability:** solutions that reduce environmental impact and build resilience.

RECALL AND CHECK

Cover the answers. (1) Give the exact definition of a value chain. (2) Name the value-chain segment that recycles waste into biofuels. (3) Is "food security is a major benefit of agripreneurship" true or false?

ANSWERS (1) The entire series of activities and transactions needed to make a product and deliver it to consumers. (2) Agricultural Waste Management. (3) True.

MODULE TWO

Fish Farming

This is the module the tests love: nearly two in three questions came from here. Fish farming rewards a farmer who reads the water and reads the fish. Learn the three water parameters, the four behaviour patterns, and the 3M wealth model, and you have the heart of the paper.



Fish Farming, Investment and Water Quality

What fish farming is, the parts of a fish farm, why it is worth investing in, and the single most tested topic in the course: the quality of the water.

What fish farming is

● MUST MEMORISE

DEFINITIONS

Aquaculture is the controlled cultivation or rearing of aquatic plants and animals in a water environment for commercial, recreational, and scientific purposes.

Fish farming is a branch of aquaculture, known as **Pisciculture**, that deals with the **science and art of rearing fish**.

◆ **SCIENCE AND ART: "ART" IS THE FILL-IN ANSWER (TEST 2 ASKED "THE SCIENCE AND _____ OF REARING FISH"). PISCICULTURE IS THE ONE-WORD NAME FOR FISH FARMING.**

PICK

PRACTICE · CF. TEST 1 Q13

What is the basic definition of fish farming given in the lecture note?

- ☒ **a The science and art of rearing fishes**
- ☐ **b The study of high population growth in Nigeria**
- ☐ **c The importation of 1.9 million metric tons of seafood**
- ☐ **d The construction of car washes and restaurants**

ANSWER **a**. Straight from the definition. The distractors are deliberately silly, the examiner's habit when the correct answer is a definition you either know or you do not.

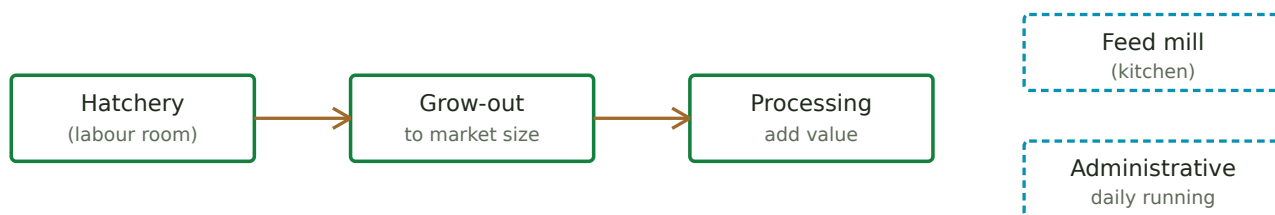
The five departments of a fish farm

● MUST MEMORISE

FIVE DEPARTMENTS

- ▶ **Hatchery** (the "labour room"): production of baby fish, the fingerlings and juveniles.
- ▶ **Grow-out**: rearing fingerlings and juveniles to mature, marketable adult size.
- ▶ **Processing**: adding value to the adult fish.
- ▶ **Feed mill** (the "kitchen"): production of the fish feed.
- ▶ **Administrative**: running the day-to-day operation of the farm.

◆ **THE TWO NICKNAMES ARE EXAM GOLD: HATCHERY = LABOUR ROOM, FEED MILL = KITCHEN. BOTH HAVE BEEN ASKED AS FILL-IN QUESTIONS.**



The three production departments in line, with the feed mill (kitchen) and the administrative unit supporting them.

FILL IN THE GAP

PRACTICE · CF. TEST 2 Q21, Q23

1. The unit of a fish farm where fingerlings and juveniles are produced is the _____ .
2. The feed-mill department is often referred to as the _____ of a fish farm.

ANSWERS 1. hatchery 2. kitchen.

Why it pays: importance and investment

TEACH

TEXT 2.1.2, 2.1.3

Fish gives high protein and low cholesterol (its fat is mostly heart-healthy Omega-3), and farming it makes fish seed available all year, easing the pressure on wild stocks. Beyond rearing, the value chain has many investment points: fish marketing, breeding (a hatchery), processing, **fish feed production** (feed is 60 to 70% of production cost), pond construction, consultancy, transport, training, **aquaponic and integrated farming**, and agri-tech.

● MUST MEMORISE

INDICES FOR EXPLOITATION

The reasons (indices) Nigeria's fish-farming landscape is worth exploiting:

- ▶ **High population**
- ▶ **High demand for fish**
- ▶ **Insufficient production of farmed fishes**
- ▶ **Abundant water and land resources**
- ▶ **Low supply of fishes**

◆ **HIGH POPULATION + HIGH DEMAND + INSUFFICIENT PRODUCTION: THE TEST ASKS WHICH COMBINATION IS THE "PRIMARY DRIVER", AND PAIRS HIGH POPULATION WITH INSUFFICIENT PRODUCTION.**

1. The high demand for fish in Nigeria is partly driven by its _____ population.
2. _____ farming combines fish production with the cultivation of plants in a linked water-based system.

ANSWERS 1. high 2. Aquaponic.

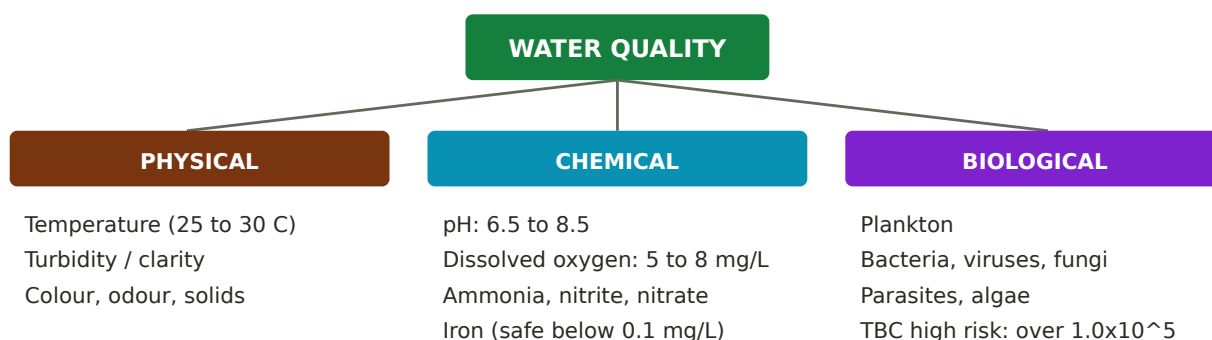
Water quality: the three parameters

● MUST MEMORISE

DEFINITION AND THE BIG THREE

Water quality is the optimal **physical, chemical, and biological** parameters of water that influence fish health, growth, and productivity: the standard of water that lets the fish thrive.

There are exactly **three** groups of parameters. Knowing which group a measurement belongs to is asked directly, so learn the members of each group cold.



Thumb rule: never venture into fish farming without knowing the quality of your water.

The three water-quality parameters and the members of each group, with the safe ranges that appear as multiple-choice answers.

TEACH

THE NUMBERS THAT GET ASKED

- ▶ **pH:** ideal **6.5 to 8.5** for most fish.
- ▶ **Dissolved oxygen (DO):** minimum **5 to 8 mg/L**. Low DO stresses and kills.
- ▶ **Ammonia:** toxic above 0.05 mg/L. **Nitrite:** keep below 0.1 mg/L. **Nitrate:** above 50 mg/L affects health. **Carbon dioxide:** above 20 mg/L suffocates.
- ▶ **Iron:** safe below 0.1 mg/L, moderate risk 0.1 to 0.5, high risk above 0.5 mg/L.
- ▶ **Total Bacterial Count (TBC):** "high risk" is above **1.0×10^5 CFU/mL**.

⚠ TRAP**PRACTICE · CF. TEST 1 Q1**

The dissolved oxygen question offers **6.5 to 8.5 mg/L** as an option. That is the **pH** range wearing a mg/L unit. The DO answer is **5 to 8 mg/L**. Two ranges, two parameters: mixing them up is the whole trap. Learn the pair together so you can never be baited: pH 6.5 to 8.5, DO 5 to 8.

PICK**PRACTICE · CF. TEST 2 Q5, Q26**

A technician measures ammonia, salinity, pH, hardness and dissolved minerals. Under which group do these fall?

- ☐ a Physical parameters
- ☒ b **Chemical parameters**
- ☐ c Operational parameters
- ☐ d Biological parameters

ANSWER **b, chemical**. Ammonia, pH and dissolved minerals are chemistry. Its partner question gives high temperature and cloudy water from suspended particles, which are **physical** (turbidity, colour, temperature, suspended solids). "Operational parameters" is not one of the three groups, so it is always wrong.

FILL IN THE GAP**PRACTICE · CF. TEST 2 Q22, Q7, Q19**

1. The stated pH range suitable for rearing fish is 6.5 to _____.
2. Regular _____ changes are recommended for maintaining good water quality in an aquarium.
3. A high total plate count is commonly associated with _____ sanitation.

NOT IN THE COURSE TEXT

ANSWERS 1. 8.5 2. water 3. poor. Note on 3: the test says "total plate count", but the course text only uses "**Total Bacterial Count (TBC)**". Same idea, different name; answer "poor" regardless, and do not go hunting for "plate count" in the text.

Aquarium management

● MUST MEMORISE**THE MAINTENANCE SCHEDULE**

- ▶ **Setup**: substrate and decorations, condition the water, seal the glass, and **know your fish species** (do not stock an aggressive fish with a timid one).
- ▶ **Daily**: feed small amounts one to two times, avoid overfeeding (it causes **water pollution**); observe the fish.
- ▶ **Weekly**: **partial water changes** (20 to 30%), check parameters, clean the glass and decorations.
- ▶ **Monthly**: filter maintenance, inspect equipment.
- ▶ **Long-term**: quarantine new fish, trim or replace plants.

An aquarist wants to avoid stocking highly aggressive or incompatible species together. Which setup step directly addresses this?

- a Know the fish species**
- b Condition the water
- c Seal the glass
- d Install substrate and decorations

ANSWER **a**. Compatibility is a species question. The others are real setup steps but address leaks, chemistry and habitat, not aggression.

RECALL AND CHECK

Cover the answers. (1) Name the three water-quality parameter groups. (2) What is the DO minimum, and the pH range? (3) How often are partial water changes done? (4) What does overfeeding cause?

ANSWERS (1) Physical, chemical, biological. (2) DO 5 to 8 mg/L; pH 6.5 to 8.5. (3) Weekly (20 to 30%). (4) Water pollution.

MODULE TWO · UNIT 2

Behaviour, the 3M Model and Marketing

Fish "talk" through behaviour; wealth is built through the 3M model; and none of it pays until the fish is marketed. All three are tested.

Reading fish behaviour

TEACH

TEXT 2.1.6

Fish express themselves through behaviour, and the test gives you a described behaviour and asks for the condition or the remedy. There are four patterns. Learn the tell-tale sign and the first remedy for each.

PATTERN	TELL-TALE SIGNS	FIRST REMEDY
Inability to eat	Slow, gasping at the surface, rejecting feed	Change the polluted water; treat sickness; correct parameters
Hungry fishes	Begging at the surface, lunging at food, swimming to you	Feed to satiation to avoid cannibalism; grade by size
Slow movement	Sluggish, near the bottom, reduced response to stimuli	Increase aeration (air stones, filters, air pumps)
Stress	Faded colour, clamped fins, hiding in a secluded corner	Maintain water quality; stable temperature; acclimatise new fish

1. A fish holds its fins clamped tight, hovers in a secluded corner, and has a faded body colour. The most likely condition is:

- (a) Chronic hunger
- (b) Natural hibernation
- (c) Acute overfeeding
- (d) **Stress**

2. Fish move sluggishly and respond little to food or movement (the "slow movement" pattern). A recommended remedy is:

- (a) Increase netting to check for parasites
- (b) Discontinue all feeding for 48 hours
- (c) **Increase aeration using air stones, filters, or air pumps**
- (d) Move all fish to a smaller, more confined tank

ANSWERS 1. **d, Stress** (clamped fins + faded colour + hiding is the signature). 2. **c, aeration**. Note the absolutes in the wrong options: "all feeding", "all fish", "a smaller, more confined tank" (which would worsen it).

The 3M of creating wealth

● MUST MEMORISE

MAKE, MANAGE, MULTIPLY

The 3M of creating wealth in fish farming is **Make, Manage, and Multiply**. Make is the **science** (the foundation); Manage and Multiply are the **art** (the business).

◆ **THREE MS. "MAKE" IS THE SCIENCE; A QUESTION ASKS WHICH OF MAKE / MOVEMENT / MONEY / MARKET IS ONE OF THE 3MS, AND THE ANSWER IS MAKE.**



The 3M roadmap: Make the wealth, Manage it so it does not shrink, then Multiply it into several streams of income.

● MUST MEMORISE

THE TEN MAKE PRINCIPLES

MAKE has ten principles, "like the fingers of your hand, working together":

- ▶ **Water, Land, Security, Location, Fisheries knowledge and manpower, Source of fingerlings/juveniles, Feed availability and quality, Market, Construction, Finance.**

◆ **THESE TEN ARE THE CONFIRMED ANSWER TO "WHICH ELEMENTS BELONG TO THE MAKE COMPONENT": LOCATION, CONSTRUCTION, FEED, WATER, LAND, TECHNICAL KNOW-HOW AND SECURITY.**

TEACH

MANAGE AND MULTIPLY

MANAGE keeps the money from shrinking: keeping records, financial prudence, communication skills, diligence, and **effective human resources management** (a code of conduct, policies, regulations, discipline, training).

MULTIPLY builds many streams: technological input, leverage and networking, the **marketing mix (7Ps)**, the **4Cs of marketing**, investment, reward mechanisms, and an integrated approach. Before all this comes research: a **SWOT** analysis and a **SMART** analysis.

● MUST MEMORISE

7PS AND 4CS

7Ps of marketing: product, promotion, place, price, physical evidence (packaging), people, process (positioning).

4Cs of marketing: customer, cost, convenience, communication. (The customer is king.)

◆ **THE "MULTIPLYING EFFECT" NEEDS THE 7PS AND 4CS TOGETHER; SWOT AND SMART ARE THE RESEARCH PAIR. DO NOT CONFUSE THE TWO PAIRS.**

1. To achieve a complete "multiplying effect", marketing must integrate which frameworks?

- ☒ a The 7Ps and 4Cs of Marketing
- ☐ b The SMART and Value Addition metrics
- ☐ c The 3Ms and SWOT Analysis
- ☐ d The USP and Hygiene Assurance protocols

2. In the "Manage" phase, a critical part of effective human resources management is:

- ☐ a Securing the source of fingerlings
- ☐ b Conducting a SWOT and SMART analysis
- ☒ c Establishing a code of conduct, policy, regulation, and discipline
- ☐ d Implementing technological inputs and networking

ANSWERS 1. a (7Ps + 4Cs is the marketing pair). 2. c (HR management is the code-of-conduct answer; a is a MAKE item, d is a MULTIPLY item, b is research). The research pair SWOT + SMART is the separate answer to Test 2's "diagnostic models" question.

Marketing fresh and processed fish

● MUST MEMORISE

MARKET, BRAND, VALUE

Marketing needs **good product quality, strong branding, and smart sales strategies**. The first step is to **identify your target market**:

- ▶ **Local consumers**: households, restaurants, hotels, supermarkets.
- ▶ **Wholesale buyers, online shoppers, and the export market**.

Build a strong brand with a **Unique Selling Point (USP)**, attractive packaging, and **Hygiene and Quality Assurance**. Add value with smoked, spiced, fillet, frozen, or marinated (barbeque) fish.

FILL IN THE GAP

PRACTICE · CF. TEST 2 Q27, Q17 · TEST 1 Q14

1. The first step in marketing fresh and processed fish is identifying the target _____.
2. Sorting and _____ of fish supports uniform feed distribution and consumption.
3. To build a strong brand, a farmer must identify a USP, use attractive packaging, and ensure _____ and Quality Assurance.

ANSWERS 1. market 2. grading 3. Hygiene.

RECALL AND CHECK

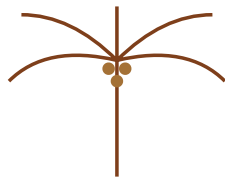
Cover the answers. (1) What are the 3Ms? (2) Give any four of the ten MAKE principles. (3) Name the 7Ps and the 4Cs. (4) What is the first step in marketing fish?

ANSWERS (1) Make, Manage, Multiply. (2) Any of: Water, Land, Security, Location, Fisheries knowledge and manpower, Source of fingerlings, Feed, Market, Construction, Finance. (3) 7Ps: product, promotion, place, price, physical evidence, people, process. 4Cs: customer, cost, convenience, communication. (4) Identify the target market.

MODULE THREE

Oil Palm

The two class tests barely touched oil palm, but the final can. This module is a bank of crisp, list-based facts, exactly what a computer-based test feeds on: the varieties, the spacing, the agronomy, and the five stages that turn a fruit bunch into oil. Learn the numbers; they are the questions.



Oil Palm Cultivation

From seed to harvest: the crop, the conditions it needs, the varieties, and the agronomy. Every underlined number here is the kind a CBT turns into a question.

● MUST MEMORISE

THE CROP AND ITS CONDITIONS

Elaeis guineensis is the oil palm. It gives palm oil (from the **mesocarp**, the fleshy part) and palm kernel oil (from the **kernel**). Optimal growing conditions:

- ▶ **Temperature:** 25 to 34 C
- ▶ **Annual rainfall:** 2000 to 2500 mm
- ▶ **Soil pH:** 5.0 to 6.0 (slightly acidic)

TEACH

TEXT 3.1.1

Cultivation runs through five stages: **propagation, nursery management, field establishment, agronomic practices, and harvesting**. The figure lays them on a timeline; the boxes below give the numbers for each.



Dura (DD) x Pisifera (PF) gives the Tenera (DP) hybrid, the variety grown commercially.

The oil palm cultivation timeline, with the numbers each stage is known by.

● MUST MEMORISE

THE NUMBERS

- ▶ **Propagation:** by seed. Break dormancy by treating seeds at **40 C for 70 to 80 days**. **Dura x Pisifera** gives the **Tenera** hybrid (highest oil yield).
- ▶ **Nursery:** seedlings in black polythene bags for **12 to 15 months**; fertiliser at 56 g per bag.
- ▶ **Field:** spacing **9 m x 9 m triangular**, giving about **143 palms per hectare**. Plant at the start of the rains.
- ▶ **Agronomy:** weeding, fertiliser **NPKMg (12:12:17:2)**, pruning, pest and disease control.
- ▶ **Harvest:** first fruit **3 to 4 years** after planting. Ready signs: loose fruits falling, colour change, softening. Tools: chisel, harvesting hook, climbing rope, mechanical harvester.

TEACH

FRUIT TYPES VS SPECIES

Two different classifications, and telling them apart is a classic exam trap. **Fruit types** describe the outside (colour and morphology). **Species** describe the inside (the shell, genetic).

FRUIT TYPES (EXTERNAL)	SPECIES (INTERNAL / GENETIC)
Nigrescens: black, turning red when ripe	Dura (DD): thick shell, no fibre ring
Virescens: green, turning reddish-orange	Pisifera (PF): thin or no shell, poor fertilisation
Albescens: white, turning pale yellow	Tenera (DP): the recommended hybrid
Mantle / Poisson; Parthenocarpic (seedless)	Tenera = Dura x Pisifera crossed

⚠ TRAP

TYPE IS NOT SPECIES

Nigrescens, Virescens and Albescens are fruit **types** (colour). Dura, Pisifera and Tenera are **species** (shell). A question that describes "black turning red when ripe" wants Nigrescens, not Dura; one that says "thick shell, no fibre ring" wants Dura, not Nigrescens. Keep the two columns apart in your head.

● MUST MEMORISE

PRODUCTS AND BY-PRODUCTS

- ▶ **Palm oil** (from the mesocarp): food, cosmetics, biofuel.
- ▶ **Palm kernel oil** (from the kernel): frying, soap, cosmetics.
- ▶ **Palm kernel cake (PKC):** livestock feed, organic fertiliser.
- ▶ **Palm pressed fibre (PPF):** biomass energy, animal bedding.
- ▶ **Empty fruit bunches (EFB):** organic fertiliser, bioenergy, biodegradable materials.

PICK

PRACTICE

Which variety of oil palm is recommended for commercial cultivation, and how is it produced?

- ☐ a Dura, produced by treating seeds at 40 C for 70 days
- ☒ b **Tenera, produced by crossing Dura and Pisifera**
- ☐ c Pisifera, produced by intercropping with cassava
- ☐ d Nigrescens, produced by pruning old fronds

ANSWER **b.** Tenera (DP) combines Dura's and Pisifera's best traits: thick mesocarp, thin shell, high oil yield. Note c and d confuse a species with a fruit type and a practice.

1. Oil palm seeds have their dormancy broken by heating at 40 C for _____ days.
2. The recommended field spacing of 9 m by 9 m in a triangular pattern gives about _____ palms per hectare.
3. Palm oil is extracted from the _____ of the fruit.

ANSWERS 1. 70 to 80 2. 143 3. mesocarp.

MODULE THREE • UNIT 2

Oil Palm Processing

Turning fresh fruit bunches into oil is five ordered steps. Learn the order and what each does; the order itself is a favourite question.

● MUST MEMORISE

THE FIVE PROCESSING STAGES, IN ORDER

1. **Sterilisation:** boil the fresh fruit bunches (FFB) with steam, about **140 to 150 C for 60 to 90 minutes**. It kills pathogens, inactivates enzymes, loosens fruits, and limits free fatty acids.
2. **Threshing:** remove the fruits from the sterilised bunch stalks.
3. **Digesting:** mill the fruits to break the cell walls and separate the mesocarp from the kernel (de-pulping).
4. **Pressing / extraction:** press the digested pulp (screw, hydraulic, or centrifugal press) to release **crude palm oil (CPO)**.
5. **Clarification:** mix the crude oil with water and heat, then skim to separate solids and sludge from the oil.



The five stages in order: sterilise, thresh, digest, press, clarify.

TEACH

KERNEL, QUALITY, BY-PRODUCTS

Palm kernel is processed separately: separate the fibre, dry the nuts (sun or mechanical), crack them, extract the oil, and store (nuts in jute bags, oil in tanks).

Oil quality depends on harvesting time, sterilisation time and temperature, pressing conditions, and storage and handling.

By-products: **EFB** (fertiliser, bioenergy), **PKC** (animal feed), and **POME**, palm oil mill effluent, a wastewater with high BOD and COD that must be treated before discharge (it can be turned into biogas or compost).

Which sequence correctly orders the processing of palm fruit into crude oil?

- (a) Threshing, sterilisation, pressing, digesting, clarification
- (b) **Sterilisation, threshing, digesting, pressing, clarification**
- (c) Digesting, pressing, sterilisation, threshing, clarification
- (d) Sterilisation, digesting, threshing, clarification, pressing

ANSWER **b**. Sterilise first (you cannot thresh a raw bunch well), then thresh, digest, press for crude palm oil, and clarify. The order is the question.

1. Sterilisation of fresh fruit bunches is done with steam at about 140 to 150 C for 60 to _____ minutes.
2. The wastewater by-product of palm oil milling, high in organic matter, is called _____.
3. The solid residue left after extracting oil from the kernel, used as animal feed, is _____.

ANSWERS 1. 90 2. POME (palm oil mill effluent) 3. PKC (palm kernel cake).

Cover the answers. (1) Name the five processing stages in order. (2) What are the three fruit types and the three species? (3) What spacing gives 143 palms per hectare? (4) What is POME and why does it matter?

ANSWERS (1) Sterilisation, threshing, digesting, pressing, clarification. (2) Types: Nigrescens, Virescens, Albescens. Species: Dura, Pisifera, Tenera. (3) 9 m x 9 m triangular. (4) Palm oil mill effluent, a high-BOD/COD wastewater that must be treated before discharge.

MODULE FOUR

Agribusiness Opportunities and Value Addition

The business end of the course. How to judge an opportunity, how the agricultural value chain adds worth at every step, the six ways to add value, and how production and operations are managed. Like oil palm, it is list-heavy and CBT-friendly, so learn the lists.



Identifying Agribusiness Opportunities

What makes an agribusiness idea worth pursuing, the chain that adds value from input to consumer, and the problems that get in the way.

● MUST MEMORISE

SIX EVALUATION CRITERIA

An **agribusiness opportunity** is a potentially profitable business idea in agriculture. Judge one against six criteria:

- ▶ **Market Demand, Cost of Inputs, Profit Margin, Scalability, Environmental and Social Impact, Policy and Regulatory Compliance.**

● MUST MEMORISE

THE AGRICULTURAL VALUE CHAIN

Each step adds value, from farm to consumer:

- ▶ **Input Supply** (seeds, fertilisers, machinery)
- ▶ **Production** (crop cultivation or livestock rearing)
- ▶ **Harvesting and Primary Processing** (sorting, grading, drying)
- ▶ **Storage and Packaging** (warehousing, cold storage)
- ▶ **Transportation and Logistics**
- ▶ **Secondary Processing / Value Addition** (milk into cheese, maize into flour)
- ▶ **Marketing and Sales, Retail and Consumption, Support Services**

TEACH

PROBLEMS AND TRENDS

Common problems in agriculture: limited market access (reliance on middlemen), post-harvest losses (poor storage and cold chains), low productivity, input supply challenges, and poor extension services.

Current trends: precision agriculture (GPS, IoT, data analytics), urban and vertical farming, water-conservation and irrigation technology, and data-driven decision making.

FILL IN THE GAP

PRACTICE

1. The major cause of post-harvest losses is poor _____ and transportation infrastructure.
2. Growing crops in stacked layers indoors under controlled conditions is called _____ farming.

ANSWERS 1. storage 2. vertical.

A business model in disguise

PICK

PRACTICE · CF. TEST 2 Q30

A commercial farm wants to use its physical assets to provide educational weekend experiences for urban families. Which business model should it explore?

- ☐ a Agricultural waste management (composting, biofuel)
- ☐ b Agritech (software, databases, marketplaces)
- ☒ c **Agritourism (visitors learn about farming, pick produce, or stay on a working farm)**
- ☐ d Food processing (raw materials into processed goods)

ANSWER **c, agritourism.** "Educational experiences for families on the farm" is the definition of agritourism. The other three are real models, just not the one described. Match the scenario to the definition.

MODULE FOUR · UNIT 2

Value Addition and Innovation

Value addition is the whole point of agripreneurship: change a product into a more valuable state. Learn the six types, the three forms, and the four kinds of innovation.

● MUST MEMORISE

DEFINITION, TYPES, FORMS

Value addition is changing or transforming a product from its original state to a more valuable one (longer shelf life, better quality, new uses), by altering its **place, time, or form**.

Six types: Form, Place, Time, Convenience, Packaging and Branding, Nutritional.

Three forms: Primary (cleaning, sorting, grading), **Secondary** (drying, milling, juicing), **Tertiary** (packaging, labelling, retailing).



The three forms of value addition, from basic farm-level handling to advanced branding and retail.

TEACH

INNOVATION

Agribusiness innovation adopts new ideas, methods, or technologies. Four kinds: **Product** (fortified flour), **Process** (solar dryers, precision farming), **Marketing** (e-commerce), and **Organisational** (cooperatives, contract farming). Drivers: technology, changing consumer preferences, competition, and government policy.

Turning fresh tomatoes into paste, and milk into cheese, is which type of value addition?

- (a) Form value addition**
- (b) Place value addition
- (c) Time value addition
- (d) Packaging value addition

ANSWER **a.** Changing the physical form is **Form**. Place = moving it somewhere more valuable; Time = storing for off-season; Packaging = presentation.

MODULE FOUR · UNIT 3

Production and Operations Management

Turning inputs into valuable outputs efficiently. Learn the difference between production and operations, and the POCD scope, which is the most likely question here.

● MUST MEMORISE

PRODUCTION, OPERATIONS, POCD

Production is growing crops and rearing animals (the *what* and *how* of creating goods). **Operations** is broader: planning, input management, processing, logistics, and quality control, managing the whole path from input to customer.

Scope, by the **POCD** framework:

- ▶ **Planning:** decide what, when, and how to produce.
- ▶ **Organising:** structure operations, labour, and logistics.
- ▶ **Controlling:** monitor performance, manage inventory, reduce waste, ensure quality.
- ▶ **Directing:** supervise and guide workers and processes.

TEACH

ADVANTAGES, LIMITS, TECHNOLOGY

Advantages: higher productivity and profit, better quality and consistency, better resource use, and the ability to scale. **Disadvantages:** unpredictable weather, labour shortages, price fluctuations, poor infrastructure, and post-harvest losses.

Technology in operations: precision agriculture (drones, sensors, GPS tractors), IoT and smart farming, farm-management software, and automated machinery.

FILL IN THE GAP

PRACTICE

1. The POCD framework stands for Planning, Organising, _____, and Directing.
2. Growing crops and rearing animals is the _____ activity, while managing the whole input-to-customer path is **operations**.

ANSWERS 1. Controlling 2. production.

RECALL AND CHECK

Cover the answers. (1) Name the six evaluation criteria for an agribusiness opportunity. (2) Name the six types and three forms of value addition. (3) What does POCD stand for? (4) Give two current trends in agriculture.

ANSWERS (1) Market demand, cost of inputs, profit margin, scalability, environmental and social impact, policy and regulatory compliance. (2) Types: form, place, time, convenience, packaging and branding, nutritional. Forms: primary, secondary, tertiary. (3) Planning, Organising, Controlling, Directing. (4) Any two of: precision agriculture, vertical farming, water conservation, data-driven decisions.

REFERENCE

The Must-Memorise List and Glossary

Everything gold from the four modules, gathered onto a few pages for the night before. If you can reproduce this section from memory, you can answer the paper. The glossary that follows defines every term the examiner might drop into a fill-in question.



The Must-Memorise Consolidation

The enumerated lists, in one place. These are where the fill-in answers and the "which of the following" options come from. Cover the right-hand side and recite.

● MUST MEMORISE

DEFINITIONS TO QUOTE

- ▶ **Agriculture:** the science, art, and practice of cultivating plants and raising animals for food, fibre, medicine.
- ▶ **Agripreneurship:** entrepreneurship in agriculture; a business mindset brought to agriculture.
- ▶ **Value chain:** the entire series of activities and transactions needed to make a product and deliver it to consumers.
- ▶ **Aquaculture:** controlled rearing of aquatic plants and animals for commercial, recreational, and scientific purposes. **Fish farming (Pisciculture):** the science and art of rearing fish.
- ▶ **Value addition:** transforming a product from its original state to a more valuable one, by altering place, time, or form.

● MUST MEMORISE

FISH FARMING (MODULE TWO)

- ▶ **Five departments:** Hatchery (labour room), Grow-out, Processing, Feed mill (kitchen), Administrative.
- ▶ **Three water parameters:** Physical (temperature, turbidity), Chemical (pH 6.5 to 8.5; DO 5 to 8 mg/L; ammonia, nitrite, nitrate, iron), Biological (plankton, bacteria such as Aeromonas and Pseudomonas; TBC high risk above 1.0×10^5 CFU/mL).
- ▶ **Four behaviour patterns:** Inability to eat, Hungry (feed to satiation), Slow movement (increase aeration), Stress (clamped fins, faded colour).
- ▶ **3Ms:** Make (science: 10 principles), Manage (art: HR, records), Multiply (art: 7Ps, 4Cs). **7Ps:** product, promotion, place, price, physical evidence, people, process. **4Cs:** customer, cost, convenience, communication. Research pair: **SWOT** and **SMART**.
- ▶ **Indices for exploitation:** high population, high demand, insufficient production, abundant water and land, low supply.

● MUST MEMORISE

OIL PALM (MODULE THREE)

- ▶ **Elaeis guineensis.** Temp 25 to 34 C, rainfall 2000 to 2500 mm, soil pH 5.0 to 6.0.
- ▶ **Dura x Pisifera gives Tenera** (commercial). Dormancy: 40 C for 70 to 80 days. Nursery 12 to 15 months. Spacing 9 m x 9 m triangular, ~143 palms/hectare. Fertiliser NPKMg (12:12:17:2). First fruit 3 to 4 years.
- ▶ **Fruit types** (external): Nigrescens, Virescens, Albescens. **Species** (internal): Dura, Pisifera, Tenera.
- ▶ **Five processing stages:** Sterilisation (140 to 150 C, 60 to 90 min), Threshing, Digesting, Pressing (crude palm oil), Clarification.
- ▶ **By-products:** EFB (fertiliser), PKC (animal feed), POME (wastewater, high BOD/COD).

● MUST MEMORISE

AGRIBUSINESS (MODULE FOUR)

- ▶ **Six evaluation criteria:** market demand, cost of inputs, profit margin, scalability, environmental and social impact, policy and regulatory compliance.
- ▶ **Six value-addition types:** form, place, time, convenience, packaging and branding, nutritional. **Three forms:** primary, secondary, tertiary.
- ▶ **Four innovation types:** product, process, marketing, organisational.
- ▶ **POCD** (operations scope): Planning, Organising, Controlling, Directing.

REFERENCE

Glossary

Every term the paper might turn into a fill-in question, defined in one line.

TERM	MEANING
Agripreneur	Someone who spots agricultural opportunities and builds profitable, sustainable ventures.
Aquaponic farming	Fish production combined with growing plants in a linked water-based system.
Biosecurity	Measures to prevent the spread of disease and pests on a farm.
Clarification	Heating and skimming crude palm oil to separate it from water and solids.
Digesting	Milling sterilised fruit to separate the mesocarp from the kernel.
Dura / Pisifera / Tenera	Oil palm species by shell: thick / thin or none / the recommended hybrid.
Eutrophication	Over-enrichment of water with nutrients, causing algal overgrowth.
Fingerlings / Juveniles	Young fish; juveniles are larger than fingerlings.
Hatchery	The fish-farm unit ("labour room") where fingerlings and juveniles are produced.
Hypoxia	A deficiency of oxygen (in water, low dissolved oxygen).
Intercropping	Growing other crops (cassava, maize) among young oil palms.

TERM	MEANING
Mesocarp	The fleshy middle layer of the oil palm fruit; source of palm oil.
Pisciculture	The branch of aquaculture dealing with rearing fish.
POME	Palm oil mill effluent: wastewater high in organic matter; must be treated.
Propagation	Reproducing oil palm, usually by seed (or tissue culture).
Satiation	Feeding fish to fullness, to avoid cannibalism.
Sterilisation	Steaming or boiling fresh fruit bunches to kill pathogens and loosen fruits.
Stocking density	The number of fish kept per unit area or volume of water.
Threshing	Removing fruits from the sterilised bunch stalks.
Triangular spacing	Planting each palm at equal distance forming an equilateral triangle.
Turbidity	The cloudiness of water from suspended particles (a physical parameter).
Value chain	The full series of activities needed to make a product and deliver it to consumers.

TEST ONE

Test One, Fully Worked

The real Test 1, reproduced exactly and worked in full. This copy was unattempted, so unlike Test 2 there is no printed key: every answer here is derived from the course text and reasoned, and the one figure that does not appear in the text is flagged. Question 4 was intentionally omitted from the test, so this is twenty-nine questions.

29 questions · answers derived

The Twenty-Nine Questions, Worked

Verbatim from the test, broken down, then answered. Answers are derived from the course text (this test carried no key), so where a question reaches outside the text it is flagged.

AS PRINTED

TEST 1 Q1

In terms of physico-chemical parameters, which of the following represents the "Ideal Range" for Dissolved Oxygen (DO) to ensure healthy fish growth?

- a. 25°C - 30°C
- b. < 0.05 mg/L
- c. 6.5 - 8.5 mg/L
- d. 5 - 8 mg/L

BREAK IT DOWN

DO minimum is 5 to 8 mg/L. Option a is a temperature range; c is the pH range (6.5 to 8.5) baited with a mg/L unit; b is the ammonia toxicity threshold.

ANSWER

D

d, 5 to 8 mg/L. Do not confuse DO (5 to 8) with pH (6.5 to 8.5).

AS PRINTED

TEST 1 Q2

Which of the following is considered a "Physical Parameter" of water quality?

- a. Turbidity, odor, and temperature
- b. Ammonia, salinity, and pH
- c. Nitrite, Nitrate, and Iron
- d. Bacteria, viruses, and algae

BREAK IT DOWN

Physical = turbidity, temperature, colour, odour. b and c are chemical; d is biological.

ANSWER

A

a, turbidity, odor, and temperature.

AS PRINTED

TEST 1 Q3

The entire series of activities and transactions needed to make a fish product and deliver it to consumers is known as a _____

BREAK IT DOWN

Fill in the gap. This is the exact definition of a value chain.

ANSWER**VALUE CHAIN**

value chain.

AS PRINTED**TEST 1 Q5**

When fish are observed "begging" for food at the surface or lunging at food as soon as it appears, but are otherwise healthy, the primary remedy to avoid cannibalism is to feed the fishes to ____ .

BREAK IT DOWN

Fill in the gap. Hungry, healthy fish are fed to fullness to stop them eating each other.

ANSWER**SATIATION**

satiation.

AS PRINTED**TEST 1 Q6**

Which component of the value chain involves recycling agricultural waste into valuable products such as biofuels and animal feeds?

- Agricultural Finance
- Agricultural Waste Management
- Sustainable and Organic Farming
- Agricultural Consulting

BREAK IT DOWN

Recycling waste into biofuels and animal feeds is the definition of waste management.

ANSWER**B**

b, Agricultural Waste Management.

AS PRINTED**TEST 1 Q7**

A fish is observed with its fins clamped tightly against its body while hovering in a secluded corner and exhibiting a faded body color. Based on the documented behavioral patterns, what is the most likely primary condition affecting the fish?

- Chronic Hunger
- Natural Hibernation
- Acute Overfeeding
- Stress

BREAK IT DOWN

Clamped fins + faded colour + hiding is the exact signature of stress.

ANSWER**D**

d, Stress.

AS PRINTED**TEST 1 Q8**

When considering the economic overview of the Nigerian fish industry, what is the specific annual deficit that must be addressed by imports?

- a. 2.7 million metric tons
- b. 800,000 metric tons
- c. 1.9 million metric tons
- d. 125.38 billion metric tons

BREAK IT DOWN

A specific figure. **NOT IN THE COURSE TEXT**: this deficit is not stated in the study text I have. The best-supported value is 1.9 million metric tons, the figure the test itself cites elsewhere (Q13's option c names "importation of 1.9 million metric tons of seafood"). Treat as likely, not certain.

ANSWER**C (LIKELY)**

c, 1.9 million metric tons (derived; confirm against the lecturer's slides if you can).

AS PRINTED**TEST 1 Q9**

In the "3M" roadmap for wealth creation, the _____ phase is where a farmer focuses on technological input, networking, and the 7P's of the marketing mix to expand the business.

BREAK IT DOWN

Fill in the gap. Technological input, networking and the 7Ps are the Multiply phase.

ANSWER**MULTIPLY**

Multiply.

AS PRINTED**TEST 1 Q10**

According to the slides, what is a necessary requirement for effectively marketing fresh and processed fish?

- a. Selling only to international export markets without local sales
- b. A combination of good product quality, strong branding, and smart sales strategies
- c. Ignoring marketing to focus solely on the rearing process
- d. Using only offline marketing channels like local markets

BREAK IT DOWN

The measured answer combines three things; a, c and d each carry an absolute (only, solely, only).

ANSWER

B

b. Good product quality, strong branding, and smart sales strategies.

AS PRINTED

TEST 1 Q11

To achieve a complete "multiplying effect" in fish wealth creation, marketing strategies must integrate which specific frameworks?

- a. The 7P's and 4C's of Marketing
- b. The SMART and Value Addition metrics
- c. The 3M's and SWOT Analysis
- d. The USP and Hygiene Assurance protocols

BREAK IT DOWN

The multiplying effect uses the 7Ps and 4Cs together (the marketing pair, distinct from SWOT/SMART).

ANSWER

A

a, the 7P's and 4C's.

AS PRINTED

TEST 1 Q12

In the context of the agricultural value chain, which activity specifically involves transforming raw matured products into industrial or value-added consumable goods through methods like milling, smoking, or canning?

- a. Production
- b. Infrastructure Development
- c. Distribution
- d. Processing

BREAK IT DOWN

Milling, smoking, canning define processing.

ANSWER

D

d, Processing.

What is the basic definition of Fish Farming provided in the lecture note?

- a. The science and art of rearing fishes
- b. The study of high population growth in Nigeria
- c. The importation of 1.9 million metric tons of seafood
- d. The construction of car washes and restaurants

BREAK IT DOWN

Straight definition. The distractors are deliberately silly.

ANSWER**A**

a, the science and art of rearing fishes.

To professionally position a fish business and develop a strong brand, a farmer must identify a Unique Selling Point (USP), use attractive packaging, and ensure _____ & Quality Assurance.

BREAK IT DOWN

Fill in the gap. The brand trio is USP, packaging, and Hygiene and Quality Assurance.

ANSWER**HYGIENE**

Hygiene.

While traditional agriculture focuses on production, _____ involves looking for creative ways to add value to agricultural products or services to make the sector more commercially viable.

BREAK IT DOWN

Fill in the gap. Finding creative ways to add value is agriprenneurship.

ANSWER**AGRIPRENEURSHIP**

agriprenneurship.

There are several "indices" for the exploitation of the Nigerian fish farming landscape. Which specific combination of factors is cited as a primary driver?

- a. Low demand for fish and abundant industrial waste
- b. Scarcity of water and high cost of transportation
- c. High supply of fishes and low demand
- d. Insufficient production of farmed fishes and high population

BREAK IT DOWN

The indices include high population and insufficient production. The others invert the drivers.

ANSWER**D**

d. Insufficient production of farmed fishes and high population.

A farmer notices fish move sluggishly, struggle to swim, and show a reduced response to external stimuli like movement or food. According to the "Slow Movement" category, which of these is a recommended remedy for this specific behavior?

- a. Increase the frequency of netting to check for physical parasites
- b. Discontinue all feeding for 48 hours regardless of satiation
- c. Increase aeration using air stones, filters, or air pumps
- d. Immediately move all fish to a smaller, more confined tank

BREAK IT DOWN

Sluggishness points to low oxygen; the remedy is more aeration. b and d carry absolutes and would worsen it.

ANSWER**C**

c, increase aeration.

Which specific branch of agriculture is responsible for developing new crop varieties to increase yield, enhance pest resistance, or improve environmental tolerance?

- a. Plant Breeding
- b. Soil Management
- c. Agricultural Engineering
- d. Agroforestry

BREAK IT DOWN

Developing new crop varieties is plant breeding, word for word.

ANSWER**A**

a, Plant Breeding.

AS PRINTED**TEST 1 Q19**

Which of the following is listed as one of the "3Ms" of creating wealth in fish farming?

- a. Make
- b. Movement
- c. Money
- d. Market

BREAK IT DOWN

The 3Ms are Make, Manage, Multiply. Only Make is offered.

ANSWER**A**

a, Make.

AS PRINTED**TEST 1 Q20**

When evaluating the biological parameter of water, what level of Total Bacterial Count (TBC) is categorized as "High Risk," indicating a high likelihood of bacterial diseases?

- a. Greater than 1.0×10^5 CFU/mL
- b. Less than 30 NTU
- c. Exactly 5.0 mg/L
- d. Between 1.0×10^3 and 1.0×10^4 CFU/mL

BREAK IT DOWN

High risk TBC is above 1.0×10^5 CFU/mL. NTU is a turbidity unit; mg/L is a chemical unit.

ANSWER**A**

a, greater than 1.0×10^5 CFU/mL.

AS PRINTED**TEST 1 Q21**

What is the "thumb rule" mentioned regarding entering the fish farming business?

- a. Always use the most expensive filtration system available
- b. Do not venture into fish farming without knowing the standard of your water
- c. Only use water that appears crystal clear to the naked eye
- d. Feed the fish four times a day to ensure rapid growth

BREAK IT DOWN

The thumb rule is stated word for word: know your water first. c is the very trap the text warns against (clear is not always safe).

ANSWER

B

b. Do not venture in without knowing the standard of your water.

AS PRINTED

TEST 1 Q22

Which of the following environmental triggers is specifically noted for causing "shock" and stress due to sudden fluctuations, rather than just being a constant poor condition?

- a. Long-term nutritional deficiency in the feed quality
- b. Gradual accumulation of agricultural waste in the pond
- c. Lack of live plants or air stones for aeration
- d. Improper acclimatization to new water conditions and sudden temperature changes

BREAK IT DOWN

"Shock" from "sudden fluctuations" points to acclimatisation and sudden temperature change; the others are gradual or constant.

ANSWER

D

d.

AS PRINTED

TEST 1 Q23

Beyond direct rearing and marketing, which specialized service area is explicitly categorized as an investment opportunity within the fish farming value chain?

- a. National agricultural policy advocacy
- b. Tourism and aesthetic purpose
- c. Industrial chemical manufacturing for water treatment
- d. Commercial deep-sea trawling operations

BREAK IT DOWN

The investment list explicitly names "Tourism and aesthetic purpose" (an agro-tourism park).

ANSWER

B

b, Tourism and aesthetic purpose.

Which of these is listed as a target market under the "Local Consumers" category?

- a. Global wholesale buyers
- b. Fish exporters and seafood distributors
- c. Households, restaurants, hotels, and supermarkets
- d. International e-commerce platforms

BREAK IT DOWN

Local consumers are households, restaurants, hotels, supermarkets. The others are wholesale, export, or online.

ANSWER**C****C.**

What is the underlying cause when fish reject feed intake while simultaneously gasping for air at the surface and moving very slowly?

- a. Excessive use of air stones and over-oxygenation
- b. Dominant fish eating all the available feed
- c. The fish have reached their maximum marketable size
- d. Polluted water quality or illness/infection

BREAK IT DOWN

Reject feed + gasp + slow is the "inability to eat" pattern, caused by polluted water or illness.

ANSWER**D****d, polluted water quality or illness/infection.**

In the "Manage" phase of the wealth creation roadmap, which of the following is identified as a critical component of Effective Human Resources Management?

- a. Securing the physical location and source of fingerlings
- b. Conducting a SWOT and SMART analysis for innovation
- c. Establishing a code of conduct, policy, regulation, and discipline
- d. Implementing technological inputs and networking

BREAK IT DOWN

HR management is the code of conduct, policy, regulation, discipline. a is a MAKE item; d is MULTIPLY; b is research.

ANSWER**C****C.****AS PRINTED****TEST 1 Q27**

Which of the following is categorized as a "Value Addition" strategy specifically intended to expand a farmer's stream of income?

- a. Offering spiced fish, fillets, or marinated (barbeque) fish
- b. Identifying local households and restaurants as target markets
- c. Utilizing WhatsApp Business and Telegram for online sales
- d. Implementing loyalty programs and seasonal promotions

BREAK IT DOWN

Value addition (processing) means offering spiced, fillet, marinated fish. b is target markets, c is online marketing, d is promotions.

ANSWER**A**

a. Spiced fish, fillets, or marinated (barbeque) fish.

AS PRINTED**TEST 1 Q28**

Which of the following is listed as a specific investment opportunity in the value chain of fish farming?

- a. Deep-sea petroleum extraction
- b. Fish feed production
- c. Industrial plastic manufacturing
- d. Specialized textile weaving

BREAK IT DOWN

Fish feed production is on the investment list (feed is 60 to 70% of cost). The others are unrelated industries.

ANSWER**B**

b, Fish feed production.

AS PRINTED**TEST 1 Q29**

Which of these is a reason (index) mentioned for exploiting the vast opportunities in Nigeria's fish farming landscape?

- a. Excessive supply of local fish
- b. High demand for fish
- c. Low human population
- d. Lack of available land resources

BREAK IT DOWN

High demand for fish is an index. The others invert the real drivers (high population, abundant land, low supply).

ANSWER

B

b, High demand for fish.

AS PRINTED

TEST 1 Q30

According to the maintenance schedule, how often should you perform partial water changes and clean the glass/decorations?

- a. Yearly
- b. Daily
- c. Weekly
- d. Monthly

BREAK IT DOWN

Partial water changes and cleaning the glass are on the weekly schedule (daily is feeding/observing; monthly is filters).

ANSWER

C

c, Weekly.

TEST TWO

Test Two, Fully Worked

This is the real Test 2, reproduced exactly as it appeared on screen and worked in full. Because a graded copy scored 30 out of 30, every answer here is confirmed, not guessed. Each question is shown three ways: the examiner's exact words (As printed), a quick Break it down, then the Answer. Sit it once with the answers covered before you read the workings.

30 / 30 • 100%

The Thirty Questions, Worked

Verbatim from the test, then broken down, then answered. The green letter is the confirmed key.

AS PRINTED

TEST 2 Q1

Water quality is central to which of the following?

- A. A successful fish-farming business.
- B. Automating every feeding operation without human supervision.
- C. Reducing the shipping weight of commercial feed.
- D. Determining national agricultural tax exemptions.

BREAK IT DOWN

Definitional recall. Water quality is the core of a fish-farming business; the other three are off-topic noise.

ANSWER

A

A, a successful fish-farming business. Water quality is repeatedly called the core to master.

AS PRINTED

TEST 2 Q2

An entrepreneur wants to understand how agriprenurship differs from traditional farming. Which option best distinguishes an agriprenurial venture from a traditional cultivation model?

- A. Agriprenurship limits all business operations strictly to primary open-field crop cultivation and excludes digital technology.
- B. Agriprenurship eliminates biological farming and focuses exclusively on speculative international financial trading.
- C. Agriprenurship applies entrepreneurial and business principles across agriculture and its value chain, including production, processing, distribution, technology and services.
- D. Agriprenurship operates entirely as a non-profit government programme and avoids commercial transactions, value addition and financial management.

BREAK IT DOWN

Three options carry an absolute (strictly, exclusively, entirely). The measured, fullest option is the answer.

ANSWER

C

C. Agriprenurship is business principles applied across the whole agricultural value chain.

Which factors support the investment potential of fish farming in Nigeria?

- A. A large population, high demand for fish, inadequate domestic production, and substantial land and water resources.
- B. Low consumer demand, an oversupply of wild fish, limited land and extensive government subsidies.
- C. High export tax credits, strict organic mandates and consumer preference for high-cholesterol food.
- D. Low population density, low market demand and a persistent feed surplus.

BREAK IT DOWN

These are the indices for exploitation: high population, high demand, insufficient production, abundant land and water.

ANSWER

A

A. The other options invert every driver (low demand, oversupply, low population).

Which diagnostic models are identified for research and planning before starting or expanding an aquaculture venture?

- A. PESTEL analysis and Porter's Five Forces.
- B. SWOT analysis and SMART analysis.
- C. The BCG Matrix and the Ansoff Matrix.
- D. The Lean Canvas and the Balanced Scorecard.

BREAK IT DOWN

The course names two research tools: SWOT and SMART. The other pairs are real business tools the text never uses.

ANSWER

B

B, SWOT and SMART. The research pair, distinct from the marketing pair (7Ps and 4Cs).

During a water-quality assessment, a technician measures ammonia, salinity, pH, hardness and dissolved minerals. Under which group do these measurements fall?

- A. Physical parameters.
- B. Chemical parameters.
- C. Operational parameters.
- D. Biological parameters.

BREAK IT DOWN

Ammonia, pH and dissolved minerals are chemistry. "Operational parameters" is not one of the three groups.

ANSWER

B

B, chemical.

AS PRINTED

TEST 2 Q6

Which statement best defines water quality in aquaculture?

- A. The adjustment of mineral salts solely to eliminate all indigenous algae.
- B. The suitability of the physical, chemical and biological characteristics of water for fish survival, growth and health.
- C. The density of hydrogen particles required to prevent every form of microbial life.
- D. The complete removal of every organic substance from water by crystallisation.

BREAK IT DOWN

A, C and D carry absolutes (solely, every, complete). B is the measured definition: the three parameter groups.

ANSWER

B

B. Physical, chemical and biological suitability for fish.

AS PRINTED

TEST 2 Q7

Regular _____ changes are recommended for maintaining good water quality in an aquarium.

BREAK IT DOWN

Fill in the gap. The text says "partial **water** changes"; the test drops "partial", but the missing word is still water.

ANSWER

WATER

water.

AS PRINTED

TEST 2 Q8

Food security is a major benefit of agripreneurship.

Select one:

- True
- False

BREAK IT DOWN

Food security is listed as one of the three benefits of agripreneurship.

ANSWER

TRUE

True.

AS PRINTED

TEST 2 Q9

An aquarist wants to avoid stocking highly aggressive or incompatible fish species together. Which aquarium-setup step directly addresses this concern?

- A. Know the fish species.
- B. Condition the water.
- C. Seal the glass.
- D. Install substrate and decorations.

BREAK IT DOWN

Compatibility is a species question. The others are real setup steps but address leaks, chemistry and habitat.

ANSWER

A

A, know the fish species.

AS PRINTED

TEST 2 Q10

Which option contains the basic preparation steps identified for setting up an aquarium?

- A. Prepare the substrate and decorations, condition the water, ensure proper glass sealing, and understand the fish species to be stocked.
- B. Culture phytoplankton, mill feed, and apply temperature shock before stocking.
- C. Paint the tank externally, conduct genetic testing, and remove all microbial life.
- D. Carry out chemical titration, replace all the water daily, and avoid learning about the fish species.

BREAK IT DOWN

The longest, most complete option lists the four real setup steps. The others include absurd or absolute actions.

ANSWER

A

A. Substrate/decorations, condition water, seal glass, know the species.

Which strategic approach is most appropriate for building a resilient and profitable fish-farming enterprise?

- A. Convert every farm exclusively to cage culture on public rivers.
- B. Ignore marketing principles and rely only on open wholesale brokers.
- C. Depend entirely on government subventions and remove staff reward mechanisms.
- D. Be creative, maximise available resources, reduce unnecessary costs and increase productive output.

BREAK IT DOWN

A, B and C carry absolutes (every/exclusively, only, entirely). D is the measured conclusion of the wealth-creation chapter.

ANSWER

D

D. Be creative, maximise resources, cut cost, increase output.

How can fish farming help to reduce pressure on vulnerable wild fish populations?

- A. By completely banning the commercial marketing and processing of adult fish.
- B. By redirecting natural rivers away from local communities and farms.
- C. By producing fingerlings, juveniles and market-size fish throughout the year, thereby reducing reliance on overexploited wild stocks.
- D. By permanently transferring all wild aquatic organisms into indoor laboratories.

BREAK IT DOWN

Farming makes fish seed available all year, easing pressure on wild stocks. The others are absurd or absolute.

ANSWER

C

C.

The high demand for fish in Nigeria is partly driven by its _____ population.

BREAK IT DOWN

Fill in the gap. High population is the first index for exploitation.

ANSWER

HIGH

high.

_____ farming combines fish production with the cultivation of plants in a linked water-based system.

BREAK IT DOWN

Fill in the gap. Fish plus plants in a linked water system is aquaponics.

ANSWER

AQUAPONIC

Aquaponic.

Agripreneurship involves bringing a _____ mindset to agriculture.

BREAK IT DOWN

Fill in the gap. The closing definition of agripreneurship: a business mindset.

ANSWER

BUSINESS

business.

Pisciculture is an aspect of aquaculture that deals with the science and _____ of rearing fish.

BREAK IT DOWN

Fill in the gap. The definition pairs science with art.

ANSWER

ART

art.

Sorting and _____ of fish is a management technique that supports uniform feed distribution and consumption.

BREAK IT DOWN

Fill in the gap. Sorting and grading by size gives uniform feeding.

ANSWER**GRADING**

grading.

AS PRINTED**TEST 2 Q18**

Modern agriculture consists of interconnected sectors rather than isolated industries. Which example best demonstrates this interconnectedness?

- A. Livestock and crop farming are linked because crop residues can feed livestock, manure can support soil health, and healthy soils contribute to crop protection.
- B. Residential architecture and cellular communication protocols.
- C. Deep-sea petroleum drilling and international tourism visa regulations.
- D. Mineral extraction and international currency trading.

BREAK IT DOWN

The textbook example of interconnection is livestock and crops. The other pairs are unrelated to agriculture.

ANSWER**A**

A.

AS PRINTED**TEST 2 Q19**

A high total plate count is commonly associated with _____ sanitation.

BREAK IT DOWN

Fill in the gap. High bacterial count means poor hygiene. **NOT IN THE COURSE TEXT**: the text says "Total Bacterial Count (TBC)", not "total plate count"; answer regardless.

ANSWER**POOR**

poor.

AS PRINTED**TEST 2 Q20**

Overfeeding and the accumulation of uneaten feed in a fish tank or pond may lead to _____ .

BREAK IT DOWN

Fill in the gap. Uneaten feed decays and fouls the water.

ANSWER**WATER POLLUTION**

water pollution.

The unit of a fish farm where fingerlings and juveniles are produced is the _____.

BREAK IT DOWN

Fill in the gap. Baby fish are produced in the hatchery (the "labour room").

ANSWER

HATCHERY

hatchery.

The stated pH range suitable for rearing fish is 6.5 to _____.

BREAK IT DOWN

Fill in the gap. The pH range is 6.5 to 8.5. (Do not confuse with DO, 5 to 8.)

ANSWER

8.5

8.5.

The feed-mill department is often referred to as the _____ of a fish farm.

BREAK IT DOWN

Fill in the gap. The feed mill makes the food, so it is the kitchen.

ANSWER

KITCHEN

kitchen.

Poor sanitation, overfeeding and the accumulation of organic waste in a pond can lead to which problem?

- A. Automatic restoration of all water-quality parameters without water management.
- B. Excessive bacterial growth, including harmful strains such as *Aeromonas* and *Pseudomonas*.
- C. Rapid deterioration of glass solely because of the microbial plate count.
- D. The complete conversion of toxic ammonia into harmless gases without treatment.

BREAK IT DOWN

Organic waste feeds harmful bacteria. A and D are impossibly convenient; C is non-sense about glass.

ANSWER

B

B. Aeromonas and Pseudomonas are the harmful strains named in the text.

AS PRINTED

TEST 2 Q25

Which statement expresses the basic water-quality rule that should guide a prospective fish farmer?

- A. Eliminate every microorganism, including beneficial bacteria, from the system.
- B. Replace all mechanical filters daily regardless of water conditions.
- C. Do not venture into fish farming without first knowing the quality of the water to be used.
- D. Use only artificial plastic plants in every aquaculture system.

BREAK IT DOWN

This is the "thumb rule" stated word for word. A, B and D carry absolutes and are wrong.

ANSWER

C

C. Know your water before you start.

AS PRINTED

TEST 2 Q26

A fish farmer observes high water temperature and cloudy water caused by suspended particles. Under which classification do these indicators fall?

- A. Chemical parameters, which include iron, ammonia, salinity, pH and hardness.
- B. Mechanical parameters, which include filter replacement, aeration rate and tank thickness.
- C. Biological parameters, which include fungi, bacteria, viruses, parasites and algae.
- D. Physical parameters, which include turbidity, colour, odour, temperature and suspended solids.

BREAK IT DOWN

Temperature and suspended particles (turbidity) are physical. "Mechanical parameters" is not a real group.

ANSWER

D

D, physical.

The first step in marketing fresh and processed fish is identifying the target _____.

BREAK IT DOWN

Fill in the gap. Marketing starts by identifying the target market.

ANSWER

MARKET

market.

Which elements belong to the MAKE component of the fish-farming wealth-creation framework?

- A. Location, construction, feed, water, land, technical know-how and security.
- B. Marketing mix, leverage, research, SWOT analysis and an integrated approach.
- C. Car wash, restaurant, hairdressing salon, boutique, cybercafé and children's play centre.
- D. Policies, regulations, discipline, training, internal controls and corporate monitoring.

BREAK IT DOWN

MAKE is the ten foundation principles. B is MULTIPLY, D is MANAGE (HR), C is unrelated businesses.

ANSWER

A

A. Location, construction, feed, water, land, know-how, security are MAKE items.

Underfeeding and overfeeding fish are good practices for fish growth and health.

Select one:

True

False

BREAK IT DOWN

Both extremes harm the fish: underfeeding starves, overfeeding pollutes the water.

ANSWER

FALSE

False.

A commercial farm wants to use its physical assets to provide educational weekend experiences for urban families. Which business model should it explore?

- A. Agricultural waste management, which focuses on composting, manure handling and the conversion of waste into biofuel.
- B. Agritech, which focuses on developing software applications, databases and digital marketplaces.
- C. Agritourism, which creates experiences in which visitors can learn about farming, pick produce or stay on a working farm.
- D. Food processing, which focuses on converting raw agricultural materials into processed goods.

BREAK IT DOWN

"Educational experiences for families on the farm" is agritourism by definition. Match the scenario to the definition.

ANSWER**C**

C, agritourism.

MOCK CBT ONE

Forty Questions, Ten Per Module

A full-length practice test: forty questions, ten from each of the four modules, in the styles the real CBT uses. These are practice questions written for this manual, not real test questions, so they are not marked As printed. Give yourself forty minutes, cover the answers at the end, and write your choices before you check.



Question Paper

Forty questions, ten per module. No notes. Answers at the end.

Module One: Agriculture and Agripreneurship (1 to 10)

1. Agriculture is defined as the science, art, and _____ of cultivating plants and raising animals. (fill)

2. Which best describes an agripreneur as against a traditional farmer?

- ☐ a Farms only for family subsistence and avoids technology
- ☐ b Applies business principles and value addition across the value chain
- ☐ c Trades only on international financial markets
- ☐ d Runs farming entirely as a non-profit scheme

3. Which value-chain component recycles agricultural waste into biofuels and animal feeds?

- ☐ a Agricultural Finance
- ☐ b Agricultural Consulting
- ☐ c Agricultural Waste Management
- ☐ d Agricultural Policy and Advocacy

4. A value chain is best defined as:

- ☐ a The single act of harvesting at the right time
- ☐ b The entire series of activities and transactions needed to make a product and deliver it to consumers
- ☐ c The removal of all middlemen from trade
- ☐ d Storing grain only to raise its price

5. Agripreneurship involves bringing a _____ mindset to agriculture. (fill)

6. Which branch of agriculture develops new crop varieties for higher yield and pest resistance?

- ☐ a Plant Breeding
- ☐ b Soil Management
- ☐ c Agricultural Engineering
- ☐ d Agroforestry

7. True or false: food security is a major benefit of agripreneurship. (true / false)

8. Which activity transforms raw matured products into value-added goods by milling, smoking, or canning?

- ☐ a Production
- ☐ b Distribution
- ☐ c Processing
- ☐ d Infrastructure Development

9. In interconnected agriculture, livestock provide _____ that enriches the soil. (fill)

10. Match each agripreneurship example to its description.

EXAMPLE	DESCRIPTION
1. Agri-tech startup	W. Visitors learn about farming on the farm
2. Agri-tourism	X. An app connecting farmers to buyers
3. Food processing	Y. Micro-loans for small-scale farmers
4. Agri-financing	Z. Turning tomatoes into sauce

Module Two: Fish Farming (11 to 20)

11. The ideal dissolved-oxygen range for healthy fish growth is:

- ☐ a 6.5 to 8.5 mg/L
- ☐ b 5 to 8 mg/L
- ☐ c Below 0.05 mg/L
- ☐ d 25 to 30 mg/L

12. The feed-mill department is nicknamed the _____ of a fish farm. (fill)

13. Ammonia, pH, salinity and hardness are measured. They belong to which group?

- ☐ a Physical parameters
- ☐ b Biological parameters
- ☐ c Chemical parameters
- ☐ d Operational parameters

14. In the 3M model, which M is the science and foundation of fish farming?

- ☐ a Make
- ☐ b Manage
- ☐ c Multiply
- ☐ d Market

15. The stated pH range suitable for rearing fish is 6.5 to _____. (fill)

16. Fish show the "slow movement" pattern. The recommended remedy is to:

- ☐ a Increase aeration with air stones, filters or air pumps
- ☐ b Move every fish to a smaller tank
- ☐ c Stop all feeding for two days
- ☐ d Net the fish repeatedly

17. True or false: underfeeding and overfeeding are both good practices for fish growth. (true / false)

18. A Total Bacterial Count categorised as "high risk" is:

- ☐ a Greater than 1.0×10^5 CFU/mL
- ☐ b Less than 30 NTU
- ☐ c Exactly 5.0 mg/L
- ☐ d Between 1.0×10^3 and 1.0×10^4 CFU/mL

19. The hatchery produces _____ and juveniles. (fill)

20. Match each fish behaviour to its first remedy.

BEHAVIOUR	FIRST REMEDY
1. Hungry fishes	W. Increase aeration
2. Slow movement	X. Feed to satiation
3. Stress	Y. Change the polluted water; treat sickness
4. Inability to eat	Z. Keep stable temperature; acclimatise new fish

Module Three: Oil Palm (21 to 30)

21. The commercially preferred Tenera oil palm is a cross of Dura and _____. (fill)

22. Oil palm seed dormancy is broken by heating at 40 C for about _____ days. (fill)

23. Which is the correct order of the five palm-oil processing stages?

- ☐ a Threshing, sterilisation, digesting, clarification, pressing
- ☐ b Sterilisation, threshing, digesting, pressing, clarification
- ☐ c Digesting, sterilisation, threshing, pressing, clarification
- ☐ d Pressing, digesting, threshing, sterilisation, clarification

24. The optimal soil pH for oil palm cultivation is:

- ☐ a 6.5 to 8.5
- ☐ b 5.0 to 6.0
- ☐ c 2.0 to 3.0
- ☐ d 8.0 to 9.0

25. Oil palm at 9 m by 9 m triangular spacing gives about _____ palms per hectare. (fill)

26. Palm oil mill effluent (POME) is best described as:

- ☐ a The solid cake fed to livestock
- ☐ b The fibrous residue used as bedding
- ☐ c A wastewater with high organic matter that must be treated before discharge
- ☐ d The refined cooking oil sold in shops

27. True or false: palm oil comes from the mesocarp and palm kernel oil from the kernel. (true / false)

28. Oil palm trees begin producing harvestable fruit about how long after planting?

- ☐ a 3 to 4 years
- ☐ b 6 months
- ☐ c 10 to 12 years
- ☐ d 1 month

29. The three oil palm fruit types are Nigrescens, Virescens, and _____. (fill)

30. Match each oil palm by-product to its use.

BY-PRODUCT	USE
1. Palm kernel cake	W. Biomass energy, animal bedding
2. Palm pressed fibre	X. Animal feed, organic fertiliser
3. Empty fruit bunches	Y. Wastewater needing treatment
4. POME	Z. Organic fertiliser, bioenergy

Module Four: Agribusiness and Value Addition (31 to 40)

31. In production and operations management, POCD stands for Planning, Organising, _____, and Directing. (fill)

32. Turning fresh tomatoes into paste, and milk into cheese, is which type of value addition?

- ☐ a Form value addition
- ☐ b Place value addition
- ☐ c Time value addition
- ☐ d Packaging value addition

33. The three forms of value addition are primary, secondary, and _____. (fill)

34. Which is one of the six criteria for evaluating an agribusiness opportunity?

- ☐ a The colour of the packaging
- ☐ b Scalability
- ☐ c The farmer's age
- ☐ d The distance to the airport

35. True or false: production and operations mean exactly the same thing in agribusiness. (true / false)

36. A farm wants to give urban families educational weekend experiences on the farm. Which model fits?

- ☐ a Agricultural waste management
- ☐ b Agritech
- ☐ c Agritourism
- ☐ d Food processing

37. The major cause of post-harvest losses is poor _____ and transportation infrastructure. (fill)

38. Which is a recognised type of agribusiness innovation?

- ☐ a Weather innovation
- ☐ b Process innovation
- ☐ c Gravity innovation
- ☐ d Currency innovation

39. Growing crops in stacked layers indoors under controlled conditions is called _____ farming. (fill)

40. Match each value-addition type to its example.

TYPE	EXAMPLE
1. Form	W. Cold-storing mangoes to sell off-season
2. Time	X. Milling wheat into flour
3. Place	Y. Vacuum-sealed branded packaging
4. Packaging	Z. Moving produce to an urban supermarket

MOCK CBT ONE

Answers

Mark yourself out of forty. Any you miss, return to the module and the gold box it came from.

1. practice
2. b
3. c
4. b
5. business
6. a
7. True
8. c
9. manure
10. 1-X, 2-W, 3-Z, 4-Y
11. b (a is the pH range)
12. kitchen
13. c
14. a
15. 8.5
16. a
17. False
18. a
19. fingerlings
20. 1-X, 2-W, 3-Z, 4-Y
21. Pisifera
22. 70 to 80
23. b
24. b
25. 143
26. c
27. True
28. a
29. Albescens
30. 1-X, 2-W, 3-Z, 4-Y
31. Controlling
32. a
33. tertiary
34. b
35. False
36. c
37. storage
38. b
39. vertical
40. 1-X, 2-W, 3-Z, 4-Y

MOCK CBT TWO

Forty More, Ten Per Module

Fresh questions, again ten from each module in the four test styles. Practice questions written for this manual. Forty minutes, cover the answers, then check.



Question Paper

Forty questions, ten per module. No notes. Answers at the end.

Module One: Agriculture and Agripreneurship (1 to 10)

1. Which statement correctly defines an agricultural value chain?

- ☐ a Harvesting a crop at the right time
- ☐ b The entire series of activities and transactions needed to make a product and deliver it to consumers
- ☐ c Removing every middleman from trade
- ☐ d Storing grain only to raise its price

2. Agriculture is the _____, art, and practice of cultivating plants and raising animals. (fill)

3. Which is listed as an importance (benefit) of agripreneurship?

- ☐ a Ending all rural employment
- ☐ b Food security
- ☐ c Reducing crop varieties
- ☐ d Ending mechanisation

4. In interconnected agriculture, crops and their residues serve as _____ for animals. (fill)

5. Which aspect of agriculture is responsible for maintaining soil fertility and preventing erosion?

- ☐ a Plant Breeding
- ☐ b Soil Management
- ☐ c Food Science and Technology
- ☐ d Agricultural Economics

6. True or false: an agripreneur takes calculated risks and may use crop insurance. (true / false)

7. Which is an example of agri-tourism?

- ☐ a An app connecting farmers to buyers
- ☐ b Micro-loans to small-scale farmers
- ☐ c A working farm where visitors pick fruit and learn about farming
- ☐ d Turning milk into cheese

8. An agripreneur uses _____ skills to develop solutions, manage risk, and build ventures. (fill)

9. Which of these is NOT a component of the agricultural value chain?

- ☐ a Processing
- ☐ b Distribution
- ☐ c Currency speculation
- ☐ d Agricultural Finance

10. Match each trait to the right person.

TRAIT	PERSON
1. Treats farming as a scalable business	W. Traditional farmer
2. Relies on inherited practice and manual labour	X. Agripreneur
3. Focuses on branding and value addition	Y. Traditional farmer
4. Sells mostly to middlemen or local markets	Z. Agripreneur

Module Two: Fish Farming (11 to 20)

11. Fish farming is the science and _____ of rearing fish. (fill)

12. Aquaculture is best defined as:

- ☐ a The controlled rearing of aquatic plants and animals for commercial, recreational, and scientific purposes
- ☐ b The study of ocean currents
- ☐ c The sale of fishing equipment only
- ☐ d The importation of frozen seafood

13. Which department rears fingerlings and juveniles to marketable adult size?

- ☐ a Hatchery
- ☐ b Grow-out
- ☐ c Feed mill
- ☐ d Administrative

14. Feed accounts for about 60 to _____ percent of fish production cost. (fill)

15. What is the "thumb rule" on entering fish farming?

- ☐ a Always buy the most expensive filters
- ☐ b Do not venture in without knowing the quality of your water
- ☐ c Only use crystal-clear water
- ☐ d Feed four times a day

16. True or false: water that is clean and clear is not always safe for rearing fish. (true / false)

17. Hungry, healthy fish beg and lunge at food. To avoid cannibalism the remedy is to feed them to:

- ☐ a Starvation
- ☐ b Satiation
- ☐ c A strict two-day fast
- ☐ d Half their usual amount

18. Partial water changes (20 to 30%) and cleaning the glass are done _____. (fill)

19. The complete "multiplying effect" in fish wealth creation integrates which pair?

- ☐ a The 7Ps and 4Cs of marketing
- ☐ b SWOT and SMART
- ☐ c The 3Ms and the USP
- ☐ d PESTEL and Porter's Five Forces

20. Match each measurement to its water-quality group.

MEASUREMENT	GROUP
1. Turbidity, temperature	W. Chemical
2. pH, ammonia, iron	X. Biological
3. Bacteria, plankton, algae	Y. Physical
4. Dissolved oxygen	Z. Chemical

Module Three: Oil Palm (21 to 30)

21. What is the botanical name of the oil palm?

- ☐ a) Elaeis guineensis
- ☐ b) Zea mays
- ☐ c) Oryza sativa
- ☐ d) Manihot esculenta

22. Sterilisation of fresh fruit bunches uses steam at about 140 to _____ C. (fill)

23. Which fruit type is black, turning red when ripe?

- ☐ a) Virescens
- ☐ b) Albescens
- ☐ c) Nigrescens
- ☐ d) Parthenocarpic

24. Oil palm seedlings are raised in polythene bags for about 12 to _____ months. (fill)

25. The fertiliser commonly recommended for oil palm is:

- ☐ a) NPKMg (12:12:17:2)
- ☐ b) Urea only
- ☐ c) Table salt
- ☐ d) Lime only

26. True or false: the Tenera variety gives a higher oil yield than Dura. (true / false)

27. Threshing in palm oil processing means:

- ☐ a) Boiling the bunches
- ☐ b) Removing the fruits from the sterilised bunch stalks
- ☐ c) Pressing out the oil
- ☐ d) Skimming the clarified oil

28. Palm oil is extracted from the _____ of the fruit. (fill)

29. Which by-product of oil palm is used as animal feed?

- ☐ a) Palm kernel cake (PKC)
- ☐ b) POME
- ☐ c) Crude palm oil
- ☐ d) Empty fruit bunches

30. Match each cultivation detail to its stage.

DETAIL	STAGE
1. 40 C for 70 to 80 days	W. Field establishment
2. 12 to 15 months in poly bags	X. Harvesting
3. 9 m x 9 m triangular spacing	Y. Propagation (dormancy)
4. First fruit at 3 to 4 years	Z. Nursery

Module Four: Agribusiness and Value Addition (31 to 40)

31. Value addition is best defined as:

- ☐ a Selling produce at the farm gate only
- ☐ b Transforming a product from its original state to a more valuable one by altering place, time, or form
- ☐ c Removing all packaging to cut costs
- ☐ d Storing produce until it spoils

32. The POCD framework is Planning, _____, Controlling, and Directing. (fill)

33. Which is one of the six types of value addition?

- ☐ a Nutritional
- ☐ b Geographical
- ☐ c Political
- ☐ d Seasonal

34. Precision agriculture uses drones, sensors, and _____ (a positioning system) for decisions. (fill)

35. True or false: investing in cold storage helps reduce post-harvest losses. (true / false)

36. Which is a common problem faced in agriculture today?

- ☐ a Too little sunlight everywhere
- ☐ b Post-harvest losses from poor storage and transport
- ☐ c An excess of trained extension officers
- ☐ d Overabundant capital for every farmer

37. Milling wheat into flour is an example of _____ value addition (light processing). (fill)

38. An agribusiness opportunity is best described as:

- ☐ a A guaranteed government grant
- ☐ b A potentially profitable business idea within the agricultural sector
- ☐ c A ban on all agricultural trade
- ☐ d A tax levied on farmers

39. Operations manage the whole _____-to-customer path, while production creates the goods. (fill)

40. Match each innovation type to its example.

INNOVATION TYPE	EXAMPLE
1. Product	W. E-commerce selling platform
2. Process	X. Fortified flour
3. Marketing	Y. Farmer cooperative
4. Organisational	Z. Solar dryers, precision farming

MOCK CBT TWO

Answers

Out of forty. Revisit the gold box behind any you miss.

1. b
2. science
3. b
4. fodder
5. b
6. True
7. c
8. entrepreneurial
9. c
10. 1-X, 2-W, 3-Z, 4-Y
11. art
12. a
13. b
14. 70
15. b
16. True
17. b
18. weekly
19. a
20. 1-Y, 2-Z, 3-X, 4-W
21. a
22. 150
23. c
24. 15
25. a
26. True
27. b
28. mesocarp
29. a
30. 1-Y, 2-Z, 3-W, 4-X
31. b
32. Organising
33. a
34. GPS
35. True
36. b
37. secondary
38. b
39. input
40. 1-X, 2-Z, 3-W, 4-Y

MOCK CBT THREE

Halfway House

*The middle test: forty more, ten per module, four styles.
Practice questions written for this manual. Forty minutes,
cover the answers, then check.*



Question Paper

Forty questions, ten per module. No notes. Answers at the end.

Module One: Agriculture and Agripreneurship (1 to 10)

1. Which is a key component of agripreneurship?

- ☐ (a) Value addition
- ☐ (b) Currency trading
- ☐ (c) Textile weaving
- ☐ (d) Petroleum refining

2. A value chain is the series of activities and transactions needed to make a product and deliver it to _____. (fill)

3. Agripreneurship is best described as:

- ☐ (a) Bringing a business mindset to agriculture
- ☐ (b) Farming only for personal food
- ☐ (c) Avoiding all technology
- ☐ (d) A government welfare programme

4. All branches of agriculture share one thing: they work with _____ organisms. (fill)

5. Which aspect of agriculture combines trees and crops for sustainable farming?

- ☐ (a) Agroforestry
- ☐ (b) Soil Management
- ☐ (c) Plant Breeding
- ☐ (d) Animal Husbandry

6. True or false: agripreneurship is simply entrepreneurship in agriculture. (true / false)

7. Which is an example of a food processing venture?

- ☐ (a) Offering micro-loans to farmers
- ☐ (b) Turning tomatoes into sauce
- ☐ (c) A farm-stay for tourists
- ☐ (d) An app linking farmers to buyers

8. Processing transforms raw products by milling, smoking, or _____. (fill)

9. Which is stated as an importance of agripreneurship?

- ☐ (a) Economic growth in rural areas
- ☐ (b) Higher food prices for all
- ☐ (c) Fewer jobs in the economy
- ☐ (d) Reduced food production

10. Match each aspect of agriculture to its job.

ASPECT	JOB
1. Plant Breeding	W. Maintaining soil fertility
2. Soil Management	X. Developing new crop varieties
3. Irrigation	Y. Processing and preserving food
4. Food Science	Z. Efficient use of water

Module Two: Fish Farming (11 to 20)

11. Which of these is a physical parameter of water quality?

- ☐ a Turbidity
- ☐ b Ammonia
- ☐ c Bacteria
- ☐ d Nitrate

12. Pisciculture deals with the science and _____ of rearing fish. (fill)

13. Overfeeding and uneaten feed in a pond may lead to:

- ☐ a Automatic water purification
- ☐ b Water pollution
- ☐ c Higher dissolved oxygen
- ☐ d Faster growth

14. The 3M of creating wealth is Make, Manage, and _____. (fill)

15. Which set lists elements of the MAKE component?

- ☐ a Location, construction, feed, water, land, security
- ☐ b Marketing mix, leverage, SWOT analysis
- ☐ c Policies, discipline, training, monitoring
- ☐ d Car wash, boutique, restaurant

16. True or false: clamped fins and a faded body colour are signs of stress in fish. (true / false)

17. Which is a reason (index) for exploiting Nigeria's fish farming landscape?

- ☐ a Low human population
- ☐ b High demand for fish
- ☐ c Oversupply of local fish
- ☐ d Lack of land

18. Sorting and _____ of fish by size supports uniform feeding. (fill)

19. The 4Cs of marketing are customer, cost, convenience, and _____. (fill)

- ☐ a Communication
- ☐ b Competition
- ☐ c Colour
- ☐ d Capital

20. Match each 3M element to its nature.

ELEMENT	NATURE
1. Make	W. Art (managing the money)
2. Manage	X. Science (the foundation)
3. Multiply	Y. Art (many revenue streams)
4. Ten principles	Z. Belong to Make

Module Three: Oil Palm (21 to 30)

21. The optimal temperature range for oil palm cultivation is:

- ☐ a 25 to 34 C
- ☐ b 5 to 10 C
- ☐ c 40 to 50 C
- ☐ d 0 to 5 C

22. The optimal annual rainfall for oil palm is 2000 to _____ mm. (fill)

23. Which processing step mixes the crude oil with water and heats it, then skims off solids?

- ☐ a Sterilisation
- ☐ b Threshing
- ☐ c Digesting
- ☐ d Clarification

24. Newly established oil palm plantations can intercrop cassava and maize during the first _____ years. (fill)

25. Which describes the Dura species of oil palm?

- ☐ a Thick shell (endocarp), no fibre ring
- ☐ b No shell, thick mesocarp
- ☐ c The recommended commercial hybrid
- ☐ d A seedless fruit

26. True or false: palm oil mill effluent (POME) must be treated before discharge. (true / false)

27. Which is a method used to dry palm nuts before cracking?

- ☐ a Sun drying
- ☐ b Deep-frying
- ☐ c Freezing
- ☐ d Fermenting

28. The oil released by pressing the digested pulp is crude palm oil, abbreviated _____ . (fill)

29. Which tools are used to harvest oil palm fruit bunches?

- ☐ a Chisel, harvesting hook, climbing rope
- ☐ b Plough and harrow only
- ☐ c Sprayer and drone only
- ☐ d Hammer and chisel only

30. Match each fruit type to its unripe colour.

FRUIT TYPE	UNRIPE COLOUR
1. Nigrescens	W. Green
2. Virescens	X. White
3. Albescens	Y. Black
4. Parthenocarpic	Z. Seedless

Module Four: Agribusiness and Value Addition (31 to 40)

31. Which is an example of primary (basic) value addition?

- ☐ a Cleaning, sorting, and grading
- ☐ b Milling wheat into flour
- ☐ c Branding and retailing
- ☐ d Advanced canning

32. The first stage of the agricultural value chain, providing seeds and fertilisers, is _____ supply. (fill)

33. Which is a current trend in agriculture?

- ☐ a Vertical farming
- ☐ b Abandoning all technology
- ☐ c Banning irrigation
- ☐ d Ending crop research

34. Tertiary value addition includes packaging, labelling, and _____. (fill)

35. True or false: scalability is one of the criteria for evaluating an agribusiness opportunity. (true / false)

36. Production in agribusiness refers to:

- ☐ a Growing crops and rearing animals
- ☐ b Only advertising the product
- ☐ c Only transporting goods
- ☐ d Only auditing the accounts

37. In the POCD framework, directing means supervising and _____ workers and processes. (fill)

38. Which is an advantage of good production and operations management?

- ☐ a Increased productivity and profitability
- ☐ b Unpredictable weather
- ☐ c Labour shortage
- ☐ d Poor road networks

39. Value addition increases value by altering a product's place, time, or _____. (fill)

40. Match each value-chain component to its activity.

COMPONENT	ACTIVITY
1. Input Supply	W. Warehousing and cold storage
2. Production	X. Providing seeds and fertilisers
3. Storage and Packaging	Y. Selling to the final consumer
4. Retail and Consumption	Z. Crop cultivation and livestock rearing

MOCK CBT THREE

Answers

Out of forty. Revisit the gold box behind any you miss.

1. a
2. consumers
3. a
4. living
5. a
6. True
7. b
8. canning
9. a
10. 1-X, 2-W, 3-Z, 4-Y
11. a
12. art
13. b
14. Multiply
15. a
16. True
17. b
18. grading
19. a
20. 1-X, 2-W, 3-Y, 4-Z
21. a
22. 2500
23. d
24. 4
25. a
26. True
27. a
28. CPO
29. a
30. 1-Y, 2-W, 3-X, 4-Z
31. a
32. Input
33. a
34. retailing
35. True
36. a
37. guiding
38. a
39. form
40. 1-X, 2-Z, 3-W, 4-Y

MOCK CBT FOUR

Nearly There

The fourth test: forty questions, ten per module, four styles. Practice questions written for this manual. Forty minutes, cover the answers, then check.



Question Paper

Forty questions, ten per module. No notes. Answers at the end.

Module One: Agriculture and Agripreneurship (1 to 10)

1. Agriculture is the _____, art, and practice of cultivating plants and raising animals. (fill)

2. Which is NOT one of the eleven aspects of agriculture?

- ☐ a Petroleum drilling
- ☐ b Animal Husbandry
- ☐ c Soil Management
- ☐ d Agroforestry

3. An agripreneur is best described as someone who:

- ☐ a Identifies agricultural opportunities and builds profitable, sustainable ventures
- ☐ b Only grows food for the family
- ☐ c Avoids all business risk
- ☐ d Works only for a government agency

4. Agripreneurship makes the agricultural sector efficient, sustainable, and commercially _____. (fill)

5. Which value-chain segment represents the interests of farmers?

- ☐ a Agricultural Policy and Advocacy
- ☐ b Agricultural Waste Management
- ☐ c Processing
- ☐ d Distribution

6. True or false: diversification is a key component of agripreneurship. (true / false)

7. Which value-chain activity uses methods like milling, smoking, and canning?

- ☐ a Production
- ☐ b Processing
- ☐ c Marketing
- ☐ d Distribution

8. In interconnected agriculture, crops and crop residues serve as _____ for animals. (fill)

9. Which is listed as an importance of agripreneurship?

- ☐ a Sustainability
- ☐ b Higher post-harvest losses
- ☐ c Fewer jobs
- ☐ d Lower food security

10. Match each value-chain segment to its activity.

SEGMENT	ACTIVITY
1. Production	W. Recycling waste into biofuels and feeds
2. Processing	X. Cultivating crops and rearing livestock
3. Waste Management	Y. Software, drones, apps
4. Agricultural Technology	Z. Freezing, canning, smoking

Module Two: Fish Farming (11 to 20)

11. The minimum dissolved-oxygen requirement for most fish is:

- ☐ a 5 to 8 mg/L
- ☐ b 6.5 to 8.5 mg/L
- ☐ c Above 50 mg/L
- ☐ d Below 0.05 mg/L

12. Aquaculture is the controlled rearing of aquatic plants and _____. (fill)

13. Which fish-farm department adds value to the adult fish?

- ☐ a Processing
- ☐ b Hatchery
- ☐ c Feed mill
- ☐ d Administrative

14. Ammonia becomes toxic to fish above _____ mg/L. (fill)

15. Fish reject feed while gasping at the surface and moving slowly. The underlying cause is:

- ☐ a Over-oxygenation
- ☐ b The fish reaching market size
- ☐ c Polluted water quality or illness
- ☐ d A dominant fish eating all the feed

16. True or false: fish feed production is a listed investment opportunity in fish farming. (true / false)

17. Which is a target market under the "Local Consumers" category?

- ☐ a Households, restaurants, hotels, and supermarkets
- ☐ b Fish exporters and distributors
- ☐ c Global wholesale buyers
- ☐ d International e-commerce platforms

18. To build a strong brand a farmer needs a USP, attractive packaging, and _____ and Quality Assurance. (fill)

19. Which diagnostic pair is used for research and planning before starting an aquaculture venture?

- ☐ a SWOT analysis and SMART analysis
- ☐ b 7Ps and 4Cs

- ☐ c BCG Matrix and Ansoff Matrix
- ☐ d PESTEL and Porter's Five Forces

20. Match each department to its job.

DEPARTMENT	JOB
1. Hatchery	W. Produces the fish feed
2. Grow-out	X. Produces fingerlings and juveniles
3. Feed mill	Y. Adds value to adult fish
4. Processing	Z. Rears juveniles to market size

Module Three: Oil Palm (21 to 30)

21. Oil palm seed dormancy is broken by treating seeds at:

- ☐ a 40 C for 70 to 80 days
- ☐ b 100 C for one hour
- ☐ c 0 C for a week
- ☐ d 25 C for a year

22. The Tenera hybrid is a cross of Dura and _____. (fill)

23. What is the purpose of sterilisation in palm fruit processing?

- ☐ a To kill pathogens, inactivate enzymes, and loosen the fruits
- ☐ b To paint the bunches
- ☐ c To freeze the fruits
- ☐ d To grind the kernels

24. Sterilisation is carried out for about 60 to _____ minutes. (fill)

25. Which of these is a fruit species (internal), not a fruit type (external)?

- ☐ a Tenera
- ☐ b Nigrescens
- ☐ c Virescens
- ☐ d Albescens

26. True or false: the Pisifera species has poor fertilisation and a high abortion ratio. (true / false)

27. Empty fruit bunches (EFB) are commonly used as:

- ☐ a Organic fertiliser and bioenergy
- ☐ b Cooking oil
- ☐ c Drinking water
- ☐ d Building cement

28. Palm kernel oil is extracted from the _____ of the fruit. (fill)

29. Which are signs that fresh fruit bunches are ready for harvest?

- ☐ a Loose fruits falling, colour change, softening
- ☐ b A strong smell of petrol
- ☐ c The tree turning blue

- ☐ d Leaves falling off entirely

30. Match each processing stage to its action.

STAGE	ACTION
1. Sterilisation	W. Separate mesocarp from kernel
2. Threshing	X. Boil the fresh fruit bunches
3. Digesting	Y. Press out the crude palm oil
4. Pressing	Z. Remove fruits from the bunch stalks

Module Four: Agribusiness and Value Addition (31 to 40)

31. Which is one of the six types of value addition?

- ☐ a Convenience
☐ b Weather
☐ c Currency
☐ d Political

32. In the value chain, secondary processing is also called _____ addition. (fill)

33. Which is an example of organisational innovation?

- ☐ a Farmer cooperatives or contract farming
☐ b Fortified flour
☐ c Solar dryers
☐ d An e-commerce platform

34. In POCD, organising means structuring operations, labour, and _____. (fill)

35. True or false: limited market access forces farmers to rely on middlemen, reducing their profit. (true / false)

36. Which is an example of convenience value addition?

- ☐ a Ready-to-eat or ready-to-cook meals
☐ b Cold-storing produce for off-season
☐ c Moving produce to a city market
☐ d Milling grain into flour

37. Market _____ is a criterion that measures consistent, growing need for a product. (fill)

38. Which is a disadvantage of production and operations management in agriculture?

- ☐ a Unpredictable weather affecting yields
☐ b Better product quality
☐ c Higher profitability
☐ d Improved efficiency

39. Agritourism lets visitors learn about farming, stay on a working farm, or _____ produce. (fill)

40. Match each value-addition type to its definition.

TYPE	DEFINITION
1. Form	W. Storing produce for when it is scarce
2. Time	X. Changing the physical form of a product
3. Place	Y. Improving presentation and market identity
4. Packaging	Z. Making a product available where it is more valuable

MOCK CBT FOUR

Answers

Out of forty. Revisit the gold box behind any you miss.

1. science
2. a
3. a
4. viable
5. a
6. True
7. b
8. fodder
9. a
10. 1-X, 2-Z, 3-W, 4-Y
11. a
12. animals
13. a
14. 0.05
15. c
16. True
17. a
18. Hygiene
19. a
20. 1-X, 2-Z, 3-W, 4-Y
21. a
22. Pisifera
23. a
24. 90
25. a
26. True
27. a
28. kernel
29. a
30. 1-X, 2-Z, 3-W, 4-Y
31. a
32. value
33. a
34. logistics
35. True
36. a
37. demand
38. a
39. pick
40. 1-X, 2-W, 3-Z, 4-Y

MOCK CBT FIVE

The Last One

*The final test: forty questions, ten per module, four styles.
Practice questions written for this manual. Clear this clean
and you are ready. Forty minutes, then check.*



Question Paper

Forty questions, ten per module. No notes. Answers at the end.

Module One: Agriculture and Agripreneurship (1 to 10)

1. The entire series of activities and transactions needed to make a product and deliver it to consumers is a:

- ☐ a Value chain
- ☐ b Supply glut
- ☐ c Price ceiling
- ☐ d Harvest festival

2. An agripreneur uses entrepreneurial skills to manage risks and build profitable, _____ ventures. (fill)

3. Which aspect of agriculture is the rearing of livestock such as cattle, poultry, and fish?

- ☐ a Animal Husbandry
- ☐ b Crop Production
- ☐ c Agroforestry
- ☐ d Plant Breeding

4. In agriculture there is a seed time and a _____ time, during which care decides the yield. (fill)

5. Which value-chain component recycles agricultural waste into biofuels and animal feeds?

- ☐ a Agricultural Waste Management
- ☐ b Agricultural Finance
- ☐ c Marketing
- ☐ d Production

6. True or false: food security is a major benefit of agripreneurship. (true / false)

7. Managing weather, pests, and price swings with insurance and diversification is which component of agripreneurship?

- ☐ a Risk Management
- ☐ b Value Addition
- ☐ c Diversification
- ☐ d Tech Integration

8. Agripreneurship is simply entrepreneurship in _____. (fill)

9. Which value-chain segment provides loans, insurance, and investment?

- ☐ a Agricultural Finance
- ☐ b Agricultural Tourism
- ☐ c Processing
- ☐ d Distribution

10. Match each example to its agripreneurship type.

EXAMPLE	TYPE
1. App linking farmers to buyers	W. Agri-tourism
2. Farm-stay for visitors	X. Agri-tech startup
3. Turning milk into cheese	Y. Agri-financing
4. Micro-loans to farmers	Z. Food processing

Module Two: Fish Farming (11 to 20)

11. Which of these is a biological parameter of water quality?

- ☐ a Bacteria and plankton
- ☐ b pH and ammonia
- ☐ c Turbidity and temperature
- ☐ d Salinity and hardness

12. Iron is considered safe in fish-farm water below _____ mg/L. (fill)

13. Which aquarium-setup step directly avoids stocking incompatible, aggressive species together?

- ☐ a Know the fish species
- ☐ b Seal the glass
- ☐ c Condition the water
- ☐ d Install decorations

14. The 7Ps are product, promotion, place, price, physical evidence, people, and _____. (fill)

15. In the "Manage" phase, effective human resources management is chiefly about:

- ☐ a A code of conduct, policy, regulation, and discipline
- ☐ b Securing the source of fingerlings
- ☐ c Technological input and networking
- ☐ d A SWOT and SMART analysis

16. True or false: overfeeding fish leads to water pollution. (true / false)

17. Which is NOT one of the three water-quality parameter groups?

- ☐ a Operational
- ☐ b Physical
- ☐ c Chemical
- ☐ d Biological

18. Fish that beg at the surface and lunge at food are _____ fishes. (fill)

19. Marketing fresh and processed fish requires good product quality, strong branding, and smart _____ strategies. (fill)

20. Match each sign to the behaviour pattern it indicates.

SIGN	PATTERN
1. Begging and lunging at food	W. Stress
2. Sluggish, near the bottom	X. Hungry fishes
3. Clamped fins, faded colour	Y. Inability to eat
4. Rejecting feed, gasping	Z. Slow movement

Module Three: Oil Palm (21 to 30)

21. Which oil palm variety is recommended for commercial cultivation?

- ☐ a Tenera
- ☐ b Dura
- ☐ c Pisifera
- ☐ d Nigrescens

22. The optimal soil pH for oil palm is 5.0 to _____. (fill)

23. Digesting palm fruit is done to:

- ☐ a Break down cell walls and separate the mesocarp from the kernel
- ☐ b Boil the bunches
- ☐ c Remove fruits from the stalks
- ☐ d Skim the clarified oil

24. Oil palm produces its first harvestable fruit 3 to _____ years after planting. (fill)

25. Which oil palm by-product is a wastewater that must be treated before discharge?

- ☐ a POME
- ☐ b Palm kernel cake
- ☐ c Empty fruit bunches
- ☐ d Palm pressed fibre

26. True or false: oil palm is a tropical perennial crop grown mainly for its fruits. (true / false)

27. After processing, palm kernel nuts are usually stored in:

- ☐ a Big jute bags
- ☐ b Glass jars
- ☐ c Plastic bottles
- ☐ d Paper envelopes

28. The recommended oil palm fertiliser NPKMg has the ratio 12:12:17:_____. (fill)

29. Which is a factor that affects crude palm oil quality?

- ☐ a Harvesting time and sterilisation conditions
- ☐ b The colour of the mill building
- ☐ c The name of the farm
- ☐ d The day of the week

30. Match each product to its source.

PRODUCT	SOURCE
1. Palm oil	W. The kernel (seed)
2. Palm kernel oil	X. The mesocarp
3. Palm kernel cake	Y. Bunches left after fruit removal
4. Empty fruit bunches	Z. Residue after kernel oil extraction

Module Four: Agribusiness and Value Addition (31 to 40)

31. The POCD framework in operations management stands for:

- ☐ a Planning, Organising, Controlling, Directing
- ☐ b Produce, Order, Cost, Deliver
- ☐ c Plant, Operate, Collect, Distribute
- ☐ d Price, Output, Credit, Demand

32. An agribusiness opportunity is a potentially profitable business idea within the _____ sector. (fill)

33. Which is a common problem faced in agriculture today?

- ☐ a Low productivity from outdated methods
- ☐ b Too many trained farmers
- ☐ c Excess storage capacity
- ☐ d Overabundant rainfall everywhere

34. Primary value addition involves simple farm-level steps: cleaning, sorting, and _____. (fill)

35. True or false: precision agriculture uses data-driven technology to reduce input waste. (true / false)

36. Which is an importance (benefit) of value addition?

- ☐ a Job creation and reduced wastage
- ☐ b Faster spoilage of produce
- ☐ c Lower profit margins
- ☐ d Loss of market access

37. Agritech focuses on developing software applications, databases, and digital _____. (fill)

38. Which is a driver of agribusiness innovation?

- ☐ a Technological advances and changing consumer preferences
- ☐ b A fixed, unchanging market
- ☐ c The absence of competition
- ☐ d A ban on new products

39. Operations covers planning, input management, processing, logistics, and quality _____. (fill)

40. Match each term to its meaning.

TERM	MEANING
1. Scalability	W. Adherence to laws and standards
2. Profit margin	X. Potential to grow output without a proportional cost rise
3. Market demand	Y. Revenue left after all costs
4. Regulatory compliance	Z. Consistent, growing need for a product

MOCK CBT FIVE

Answers

Out of forty. A clean sweep means you are ready for the real thing.

1. a
2. sustainable
3. a
4. harvest
5. a
6. True
7. a
8. agriculture
9. a
10. 1-X, 2-W, 3-Z, 4-Y
11. a
12. 0.1
13. a
14. process
15. a
16. True
17. a
18. hungry
19. sales
20. 1-X, 2-Z, 3-W, 4-Y
21. a
22. 6.0
23. a
24. 4
25. a
26. True
27. a
28. 2
29. a
30. 1-X, 2-W, 3-Z, 4-Y
31. a
32. agricultural
33. a
34. grading
35. True
36. a
37. marketplaces
38. a
39. control
40. 1-X, 2-Y, 3-Z, 4-W